

# **MILESTONE**

**1**

**BY  
Y. M. INC.**

# Pesco, inc

Phone: 843-767-8643  
Fax: 843-767-9181

## REQUEST FOR INFORMATION SYSTEM SERVICES

| DATE OF REQUEST    | SERVICE REQUESTED FOR DEPARTMENT(S) |
|--------------------|-------------------------------------|
| September 13, 2000 | Prop DOC                            |

| SUBMITTED BY (key user contact)                                                                        | EXECUTIVE SPONSOR (funding authority)                                                                     |
|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Name</b> Yon Meyer<br><b>Title</b> Manager<br><b>Office</b> Charleston, SC<br><b>Phone</b> 767-8643 | <b>Name</b> Mike Meyer<br><b>Title</b> President<br><b>Office</b> Charleston, SC<br><b>Phone</b> 767-8643 |

### TYPE OF SERVICE REQUESTED:

- |                                                                 |                                                                         |
|-----------------------------------------------------------------|-------------------------------------------------------------------------|
| <input type="checkbox"/> Information Strategy Planning          | <input type="checkbox"/> Existing Application Enhancement               |
| <input type="checkbox"/> Business Process Analysis and Redesign | <input type="checkbox"/> Existing Application Maintenance (problem fix) |
| <input checked="" type="checkbox"/> New Application Development | <input type="checkbox"/> Not Sure                                       |
| <input type="checkbox"/> Other (please specify) _____           |                                                                         |

### BRIEF STATEMENT OF PROBLEM, OPPORTUNITY, OR DIRECTIVE (attach additional documentation as necessary)

We need software to interface with encoders on 3 axis. The software needs to read operator switch states every .5 second. It also needs to convert data and fill out a variety of reports. The data is to be then stored in a file such that the user can use it to complete future reports.

### BRIEF STATEMENT OF EXPECTED SOLUTION

We expect the software package to be installed automatically in Windows 95, 98, or 2000. The scheme must include the ability to update an installed software through e-mailed files.

### ACTION (ISS Office Use Only)

☐ Feasibility assessment approved

Assigned to \_\_\_\_\_

☐ Feasibility assessment waived

Approved Budget \$ \_\_\_\_\_

Start Date \_\_\_\_\_ Deadline \_\_\_\_\_

☐ Request delayed

Backlogged until date: \_\_\_\_\_

☐ Request rejected

Reason: \_\_\_\_\_

Authorized Signatures:

\_\_\_\_\_  
Chair, ISS Executive Steering Body

\_\_\_\_\_  
Project Executive Sponsor

## PROBLEM STATEMENTS

|                              |                                   |
|------------------------------|-----------------------------------|
| <b>PROJECT:</b> Prop DOC     | <b>PROJECT MANAGER:</b> Yon Meyer |
| <b>CREATED BY:</b> Yon Meyer | <b>LAST UPDATED BY:</b> Yon Meyer |
| <b>DATE CREATED:</b> 9/13/00 | <b>DATE LAST UPDATED:</b> 9/18/00 |

| Brief Statements of Problem, Opportunity, or Directive | Urgency  | Visibility | Annual Benefits | Priority or Rank | Proposed Solution |
|--------------------------------------------------------|----------|------------|-----------------|------------------|-------------------|
| 1. Software to interface with encoders on 3 axis       | 8months  | High       | Unknown         | 1                | New Development   |
| 2. Read operator switch states every .5 second         | 8 month  | High       | Unknown         | 1                | New development   |
| 3. it needs to convert data and fill out reports       | 8 months | med        | Unknown         | 2                | New development   |
| 4. store data for future reports                       | 8 months | med        | Unknown         | 2                | New development   |
| 5.                                                     |          |            |                 |                  |                   |
| 6.                                                     |          |            |                 |                  |                   |
| 7.                                                     |          |            |                 |                  |                   |
| 8.                                                     |          |            |                 |                  |                   |

**FIGURE 4.8**

# **MILESTONE**

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# **Project Feasibility Assessment Report**

## **SECTION I. Executive Summary**

The president of Pesco Incorporated recommended that Y. M. Inc. design the software for the company's newest product. The new product will provide Pesco's customers with a more accurate and efficient way to measure marine propellers.

The section I of this report contains the executive summary. Section II contains the background information of the project to date. Section III contains detailed recommendation.

## **SECTION II. Background Information**

The request is for, Yon Meyer at Y. M. Inc., to develop software to generate a report with data collected from new product. The software should interface with encoders on three axis, and read the operator switch states every half of a second. Furthermore, it needs to convert the data, create reports, and store data for future reports.

The activity conducted, to date, to accomplish the survey phase was to obtain a brief description of the project. More meetings with president of Pesco, Inc. are in order to further understanding of the project.

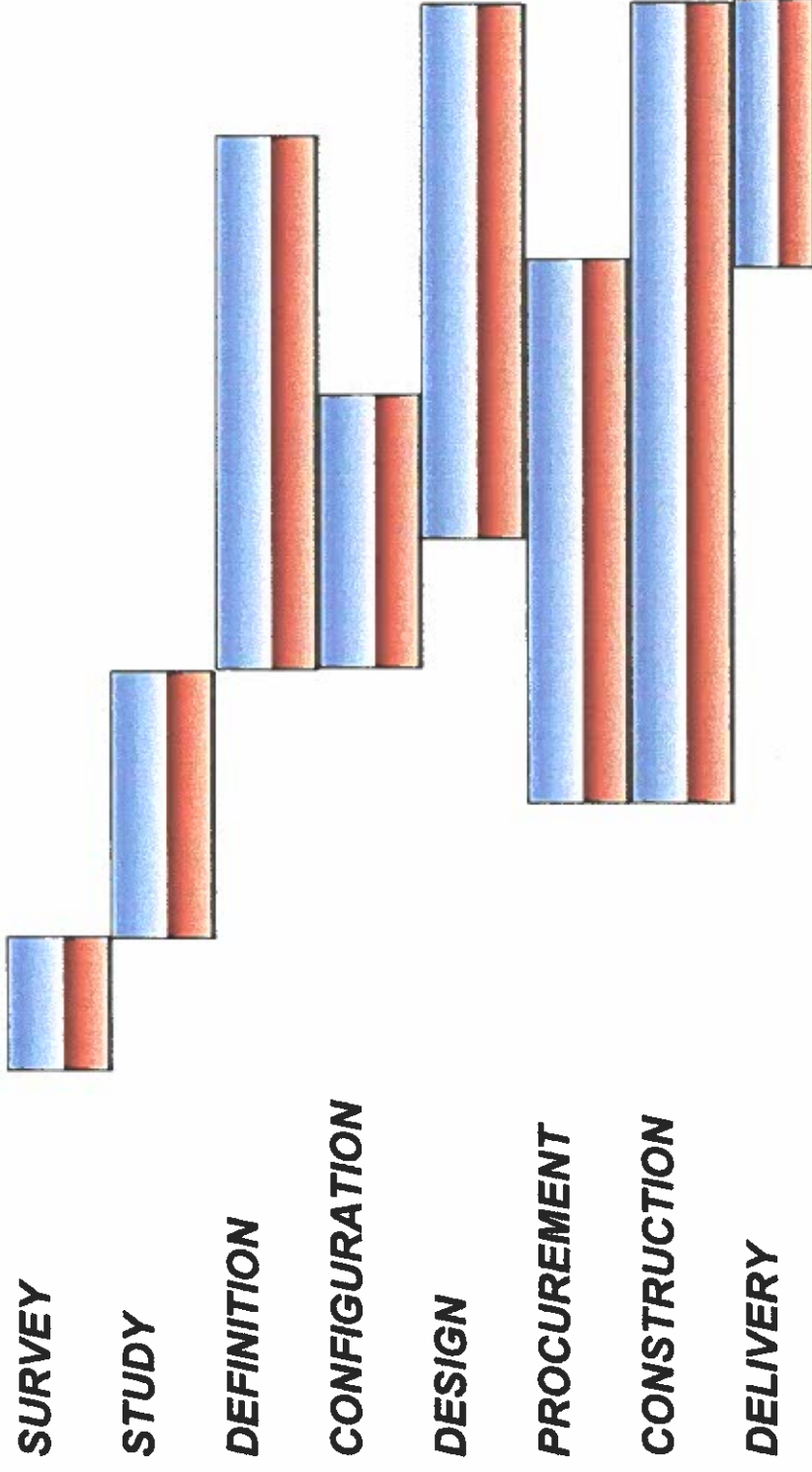
## **SECTION III. Detailed Recommendation**

It is recommended that a model of the new product be available as soon as possible. It is also recommended that the more detailed explanation of expected feature on this software. For example, what kind of report should it generate? Would this software need to generate graphics?

The project objective is to replace current measuring system and hand written report system. This replacement would add various benefits such as saving time and accuracy of the reports.

**SYSTEM DEVELOPMENT LIFE CYCLE**

**SEP OCT NOV DEC JAN FEB MAR APRIL**



# **MILESTONE**

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## PROBLEMS, OPPORTUNITIES, OBJECTIVES AND CONSTRAINTS MATRIX

|                                         |                                              |
|-----------------------------------------|----------------------------------------------|
| <b>Project:</b> Prop DOC                | <b>Project Manager:</b> Yon Meyer            |
| <b>Created by:</b> Y. M. Inc.           | <b>Last Updated by:</b> Yon Meyer            |
| <b>Date Created:</b> September 18, 2000 | <b>Date Last Updated:</b> September 24, 2000 |

| CAUSE AND EFFECT ANALYSIS                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                 | OBJECTIVES SYSTEM IMPROVEMENT                                                                                                                                                                                                                                                                                              |                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Problem or Opportunity                                                                                                                                                                                                                                                                                         | Causes and Effects                                                                                                                                                                                                                                                                                                                              | System Objective                                                                                                                                                                                                                                                                                                           | System Constraint                                                                                                                       |
| <ol style="list-style-type: none"> <li>1. With current system, the reports are generated by hand. This take too much time.</li> <li>2. There are not many software available for the marine propeller repair system.</li> <li>3. There are two available at this time, but they are very expensive.</li> </ol> | <ol style="list-style-type: none"> <li>1. Generating reports with computer saves time and effort.</li> <li>2. Current systems that are on market are too expensive and not efficient.</li> <li>3. This software will provide more accurate data Also, it will be less expensive than the other systems that are currently available.</li> </ol> | <ol style="list-style-type: none"> <li>1. Needs software to interface with encoders on 3 axis, and read operator switch states every <math>\frac{1}{2}</math> seconds.</li> <li>2. It needs to convert data, fill out report, and store data for future reports.</li> <li>3. The software is updateable through</li> </ol> | <ol style="list-style-type: none"> <li>1. The customers who do not have e-mail address won't be able to get updateable file.</li> </ol> |



# **MILESTONE**

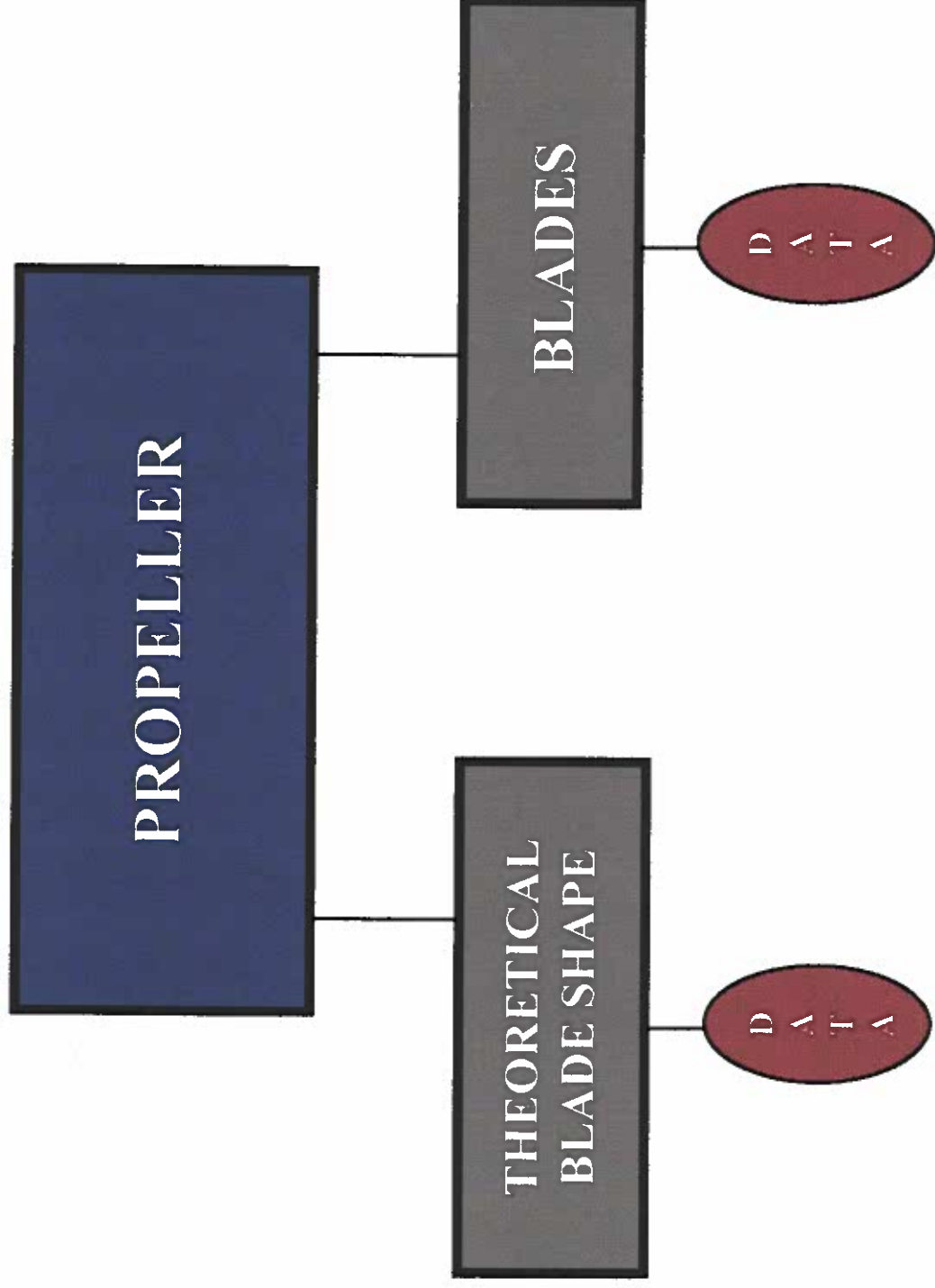
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## ENTITY/DEFINITION MATRIX

| ENTITY                         | BUSINESS DEFINITION                                                                                                                                                                                                                               |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Propeller</b>               | A propeller is a structure with helical blades to move water axially when rotated about that axis.                                                                                                                                                |
| <b>Theoretical Blade Shape</b> | <p>The theoretical blade shape is a model blade shape that is set to International Standards Organization standard.</p> <p>The other blades that are damaged are to match to these standards. Each style will have its own theoretical blade.</p> |
| <b>Blades</b>                  | Blades are the number of wings that are on propellers. It varies from propeller to propeller. Each blade will produce its own data information, and compared to the theoretical blade.                                                            |
|                                |                                                                                                                                                                                                                                                   |
|                                |                                                                                                                                                                                                                                                   |
|                                |                                                                                                                                                                                                                                                   |

# ENTITY DIAGRAM

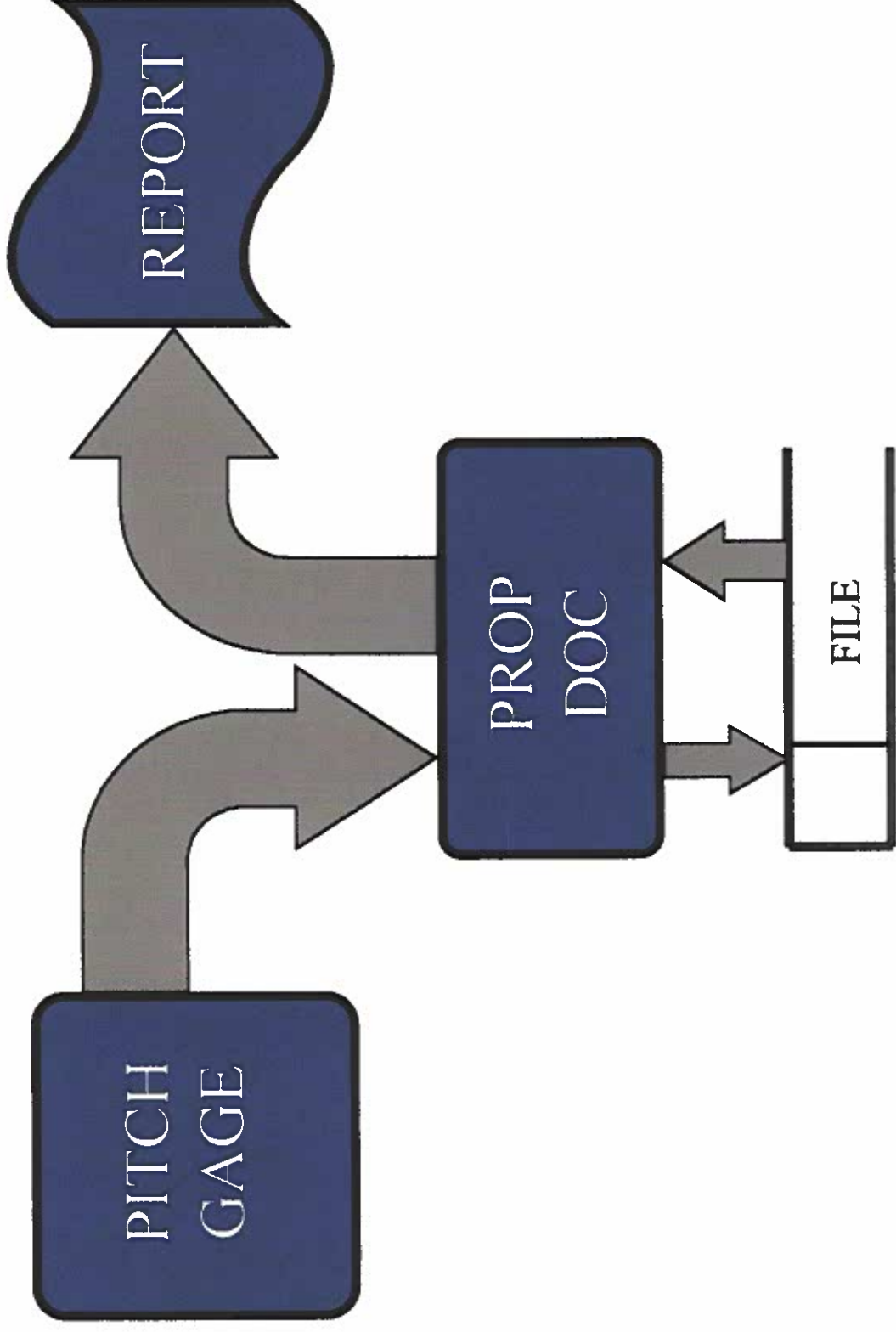


# **MILESTONE**

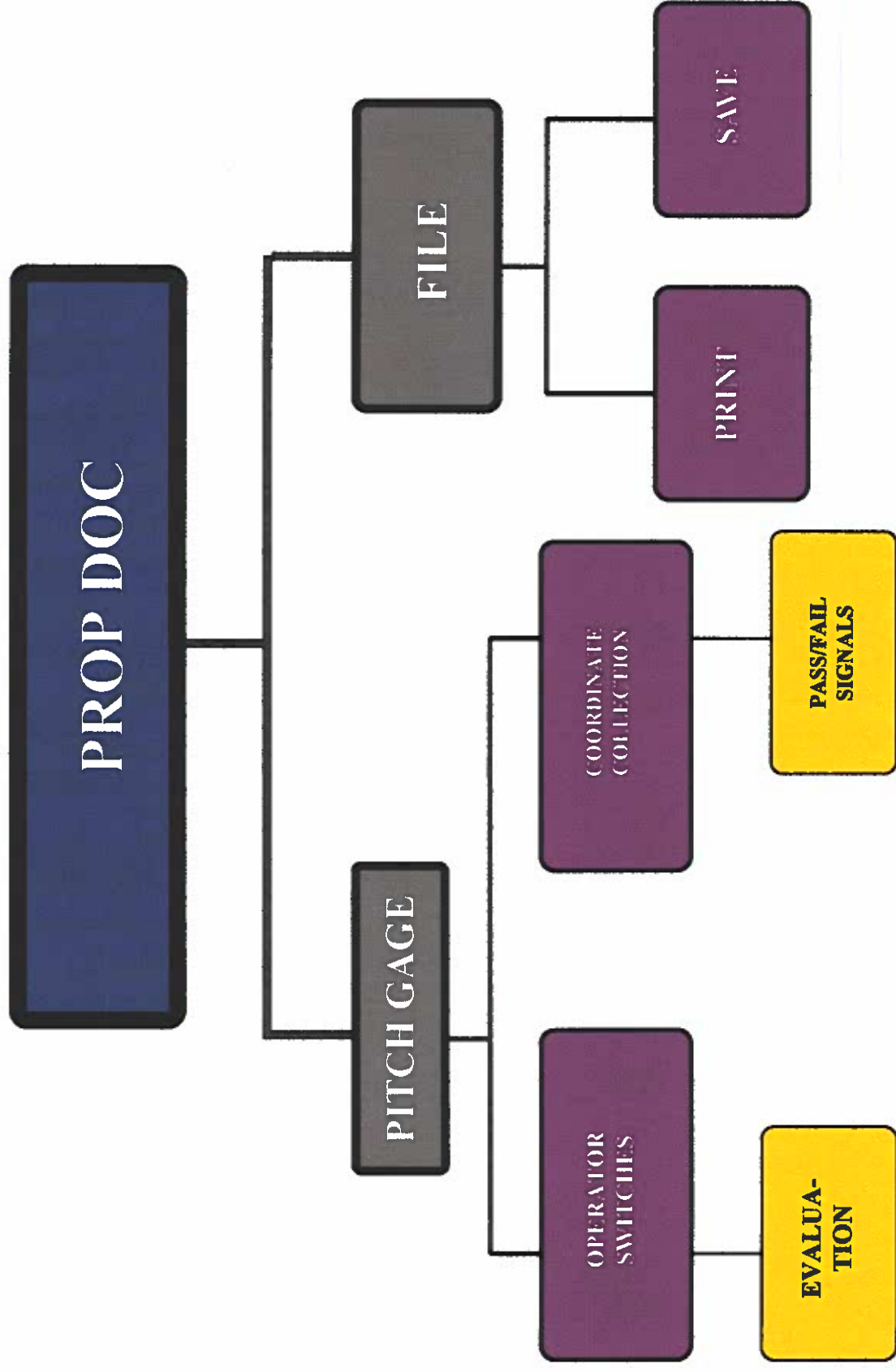
**5**

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# CONTEXT DIAGRAM

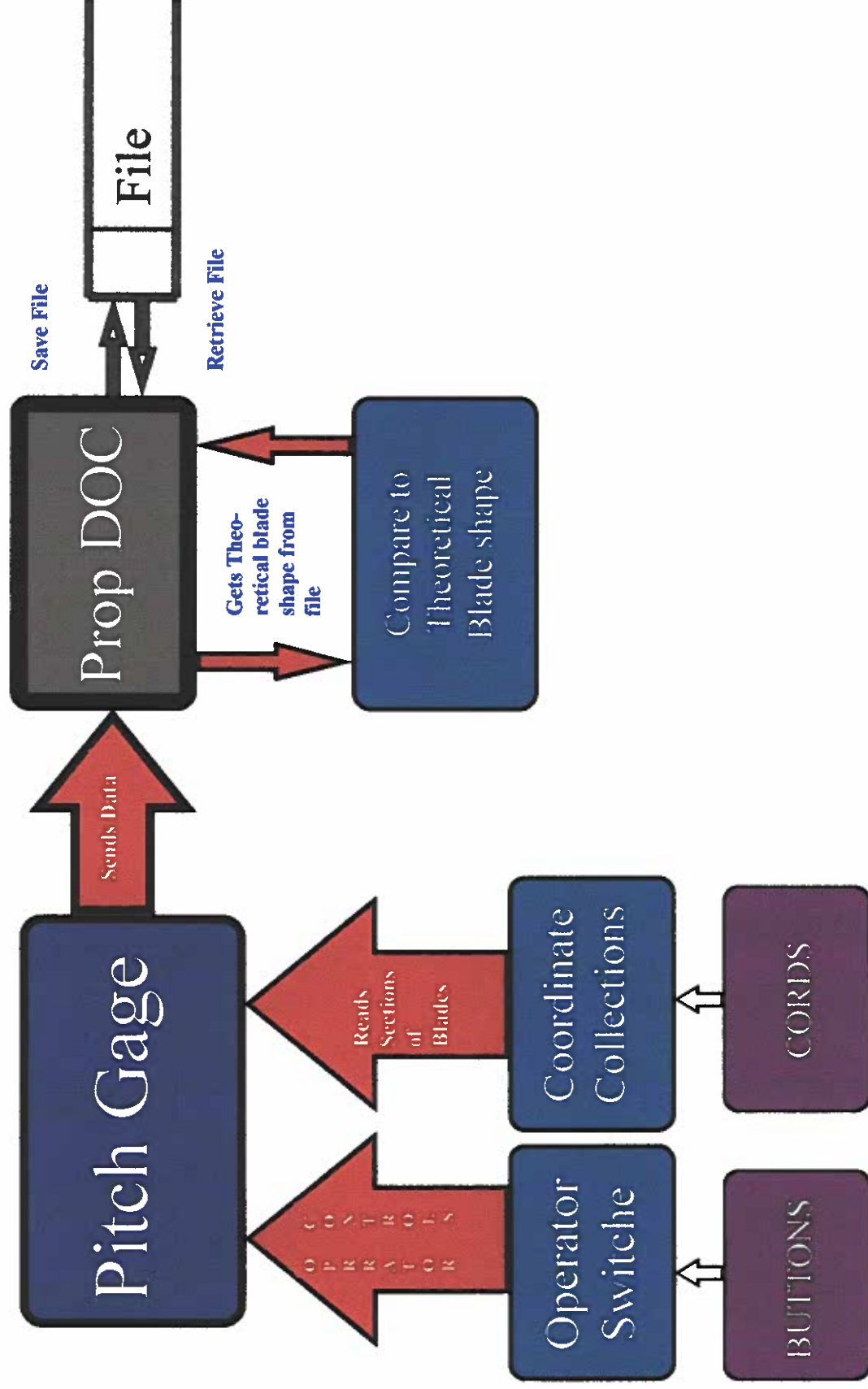


# EVENT DECOMPOSITION DIAGRAM



| Events                | Event Description                                                                          | Trigger (Input)                     | Response (Output)                                                                                                                                                                                          |
|-----------------------|--------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operator switches     | Zero each coordinates Z, $\theta$ , r                                                      | High voltage from returning switch. | Zero register for Z-axis, $\theta$ , r.                                                                                                                                                                    |
| Coordinate collection | Collecting data switch                                                                     | High voltage from returning switch. | Start storing data                                                                                                                                                                                         |
| Coordinate collection | Stop collecting data                                                                       | Low voltage from returning switch.  | Stop storing data and evaluate. If data meets its class restriction, then go to next cord. If not, then the screen will show the operator the problem.                                                     |
| Pass/fail signals     | If Passed, signal the operator to go to next cord.                                         | Response from stop collecting data. | The screen will tell the operator if failed or not.                                                                                                                                                        |
| Pass/fail signals     | If failed, signal the operator to go back to improve until it meets the class restriction. | Response from stop collecting data. | The screen will show the operator what needs to be corrected.                                                                                                                                              |
| Pass/fail signals     | Go to next cord.                                                                           | High voltage from returning switch. | It should show the operator what radius he supposed to go to.                                                                                                                                              |
| Evaluation            | Evaluate Blade/ Next Blade                                                                 | Switch                              | Computer gathers all the data from current blade, and it evaluate to see if it meets the class restriction. If yes, then go to next blade. If no, then it will indicate the problem.                       |
| evaluation            | Evaluate propeller                                                                         | Switch                              | Computer gathers all the data from current propeller, and it evaluate to see if it meets the class restriction. If yes, then it will produce certified document. If no, then it will indicate the problem. |
|                       | Save                                                                                       | Mouse button                        | File is saved.                                                                                                                                                                                             |
|                       | Print                                                                                      | Mouse button                        | Report is printed.                                                                                                                                                                                         |

# SYSTEM DIAGRAM





# DECISION TABLE 1

[illegible]

# DECISION TABLE 2

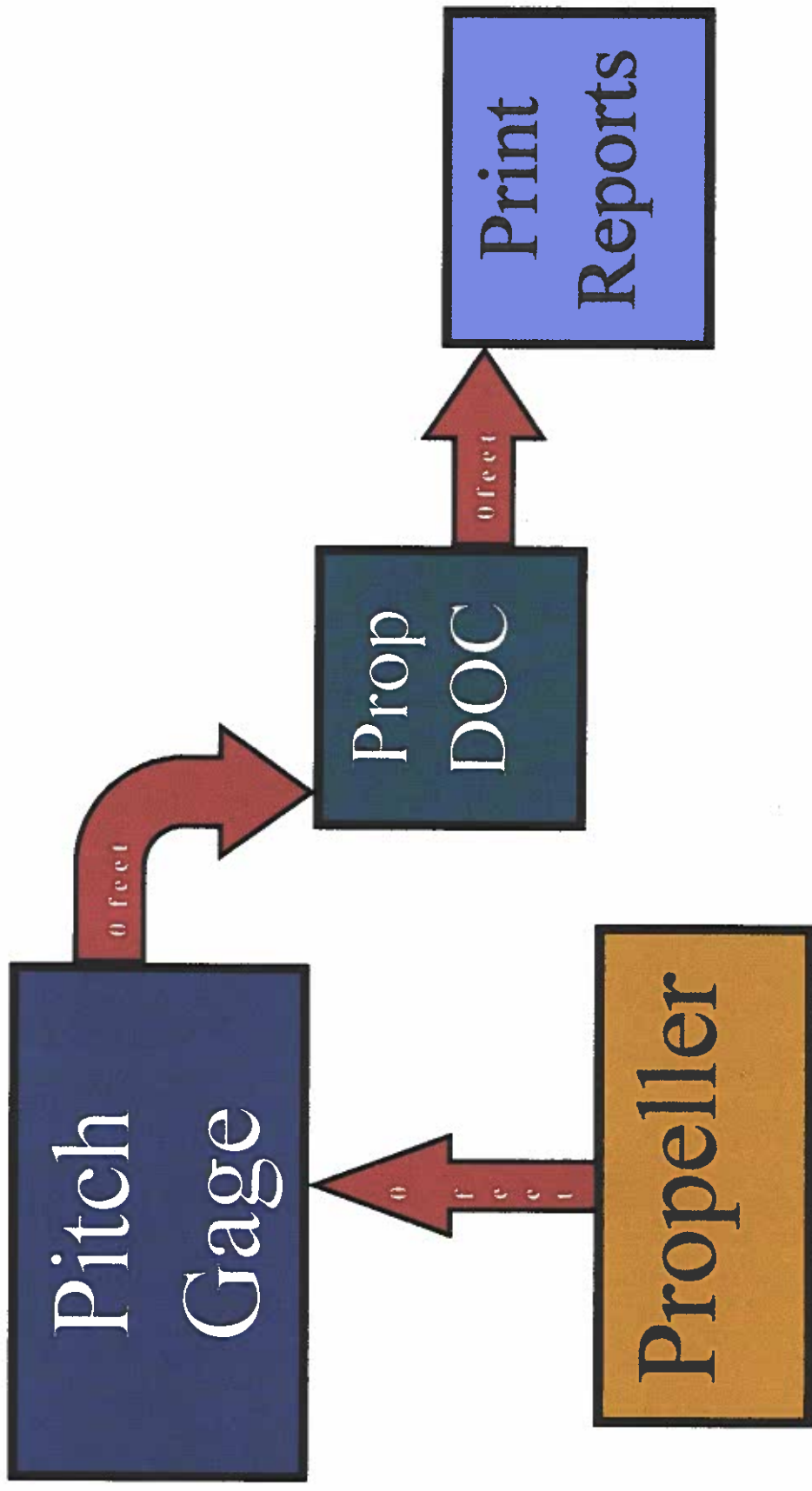
[illegible]

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# HIGH-LEVEL CONNECTIVITY DIAGRAM



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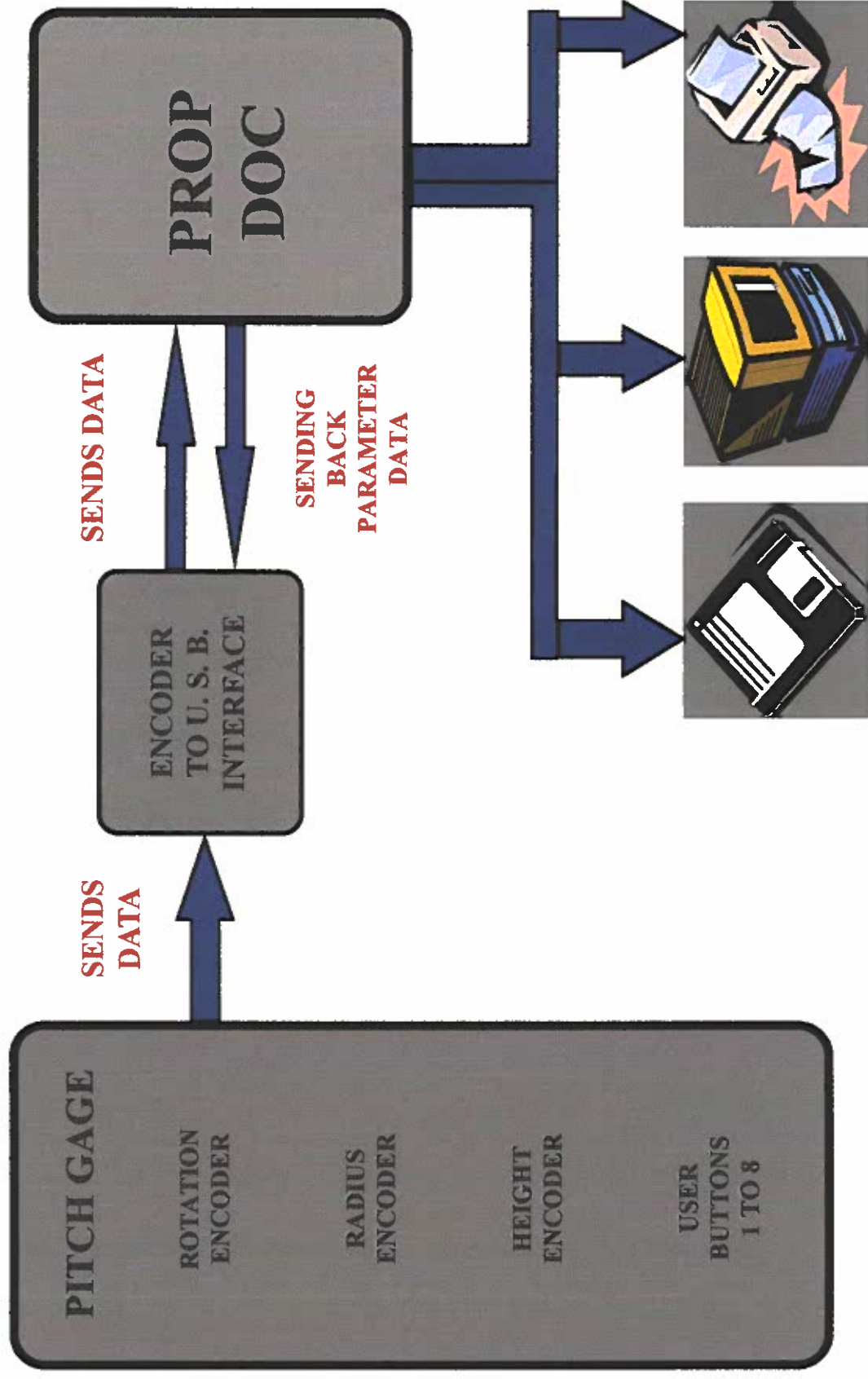
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# PHYSICAL DATA FLOW DIAGRAM



# **MILESTONE**

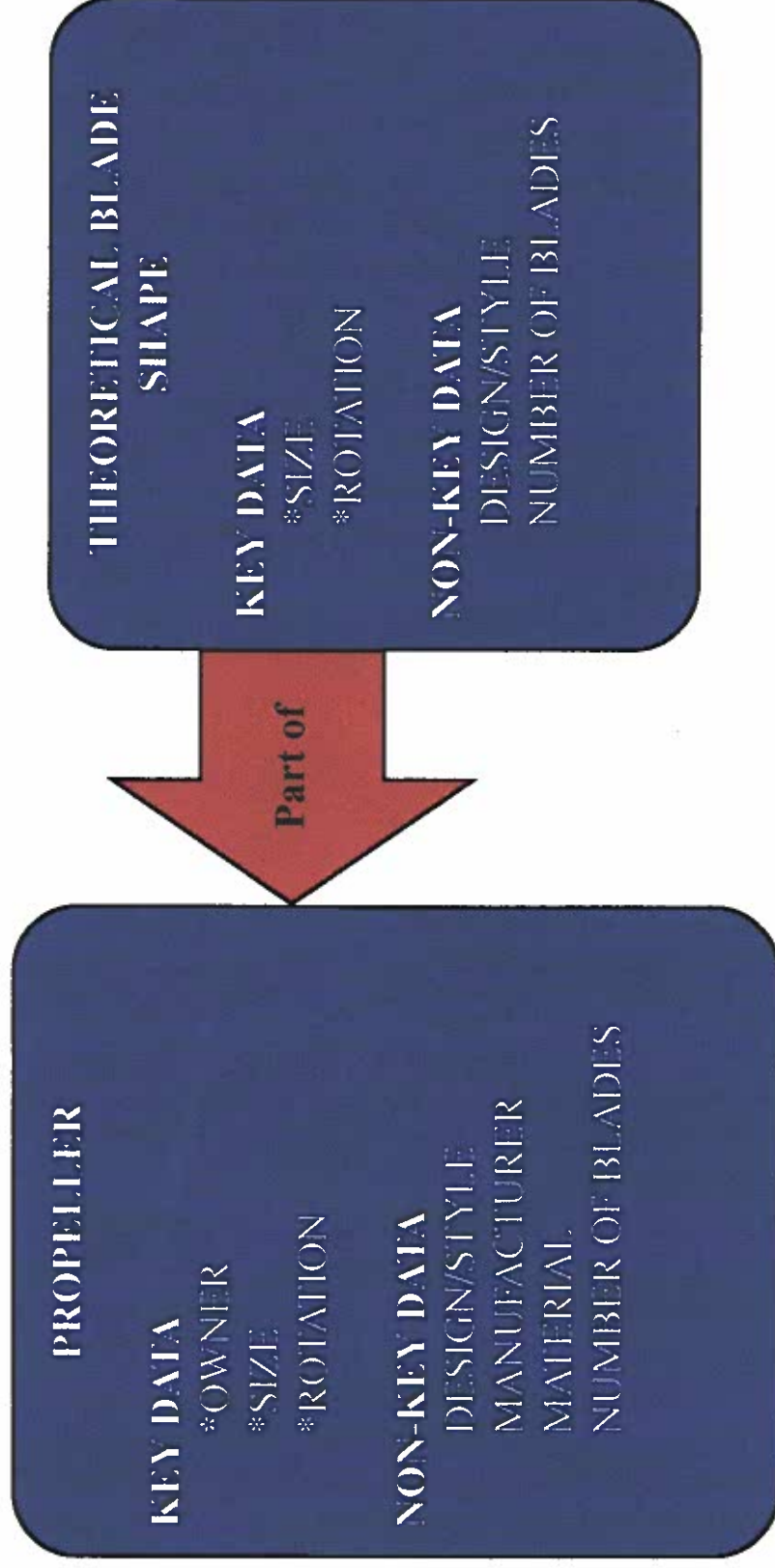
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# Logical Data Model In 3rd Normal Form

\* PRIMARY KEY



# COMPLETE TABLE DEFINITION

## PROPELLER

|                  |                    |
|------------------|--------------------|
| OWNER_ID         | VARCHAR (20) PK(1) |
| SIZE_ID          | INT PK(2)          |
| ROTATION_ID      | CHAR (1) PK(3)     |
| DESIGN/STYLE     | CHAR(10)           |
| MANUFACTURER     | CHAR(15)           |
| MATERIAL         | CHAR(8)            |
| NUMBER OF BLADES | INT                |

## THEORETICAL BLADE SHAPE

|                  |               |
|------------------|---------------|
| SIZE_ID          | INT PK(1)     |
| ROTATION_ID      | CHAR(1) PK(2) |
| DESIGN/STYLE     | CHAR(10)      |
| NUMBER OF BLADES | INT           |

# **MILESTONE**

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# Prop D O C

Enter the following items for each propellers. Use tab or arrow keys to move from item to item.

|                  |                    |
|------------------|--------------------|
| Date:            | File name:         |
| Customer Name:   | Vessel Name:       |
| Manufacturer:    | Material:          |
| Design/Style:    | Bore:              |
| Class:           | DAR:               |
| Cupping:         | Rotation:          |
| Marked Diameter: | Measured Diameter: |
| Marked Pitch:    | Measured Pitch:    |

Push enter when you are finished.

# Prop D O C

Enter the following items for each propellers. Use tab or arrow keys to move from item to item.

|                              |                          |
|------------------------------|--------------------------|
| Date: 11-10-00               | File name: ScanData1.txt |
| Customer Name: John Doe      | Vessel Name: Fish Killer |
| Manufacturer: Michigan Wheel | Material: Nibral         |
| Design/Style: Dyna Jet       | Bore: 2.0                |
| Class: I                     | DAR: .55                 |
| Cupping: 2 --.033            | Rotation: R              |
| Marked Diameter: 28          | Measured Diameter: 28.00 |
| Marked Pitch: 32             | Measured Pitch: 31.856   |

Push enter when you are finished.

|                              |                          |
|------------------------------|--------------------------|
| Date: 11-10-00               | File name: ScanData1.txt |
| Customer Name: John Doe      | Vessel Name: Fish Killer |
| Manufacturer: Michigan Wheel | Material: Nibral         |
| Design/Style: Dyna Jet       | Bore: 2.0                |
| Class: I                     | DAR: .55                 |
| Cupping: 2 -.033             | Rotation: R              |
| Marked Diameter: 28          | Measured Diameter: 28.00 |
| Marked Pitch: 32             | Measured Pitch: 31.856   |

Push enter when you are finished.

# Prop D O C

Enter the following items for each propellers. Use tab or arrow keys to move from item to item.

|                                             |                                           |
|---------------------------------------------|-------------------------------------------|
| Date: xxxxxxxxxxxx (1)                      | File Name: xxxxxxxxxxxx.xxx (2)           |
| Customer Name: xxxxxxxxxxxxxxxxxxxxxxxx (3) | Vessel Name: xxxxxxxxxxxxxxxxxxxxxxxx (4) |
| Manufacturer: xxxxxxxxxxxxxxxxxxxxxxxx (5)  | Material: xxxxxxxx (6)                    |
| Design/Style: xxxxxxxxxxxx (7)              | Bore: xxxxx (8)                           |
| Class: xx (9)                               | DAR: xxxx (10)                            |
| Cupping: xxxxxxxx (11)                      | Rotation: x (12)                          |
| Marked Diameter: xx (13)                    | Measured Diameter: xxxxxxxx (14)          |
| Marked Pitch: xx (15)                       | Measured Pitch: xxxxxxxx (16)             |

Push enter when you are finished.

## PROCESSING

# ZERO

## Select Blade



## Ready to Scan

EXIT



## STOP

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.



## Internal Report

The processing (green) light will be on as soon as ready to scan button is pushed. The stop (red) light will be on as soon as the user stops scanning.

# Net Present value Analysisi for Prop DOC

| Cash flow description                     | Year 0         | Year 1        | Year 2        | Year 3        | Year 4        | Year 5        | Total          |
|-------------------------------------------|----------------|---------------|---------------|---------------|---------------|---------------|----------------|
| Development cost:                         | \$ (20,000.00) |               |               |               |               |               |                |
| Operation & Maintenance cost:             |                | \$ (5,000.00) | \$ (5,000.00) | \$ (5,000.00) | \$ (5,000.00) | \$ (5,000.00) |                |
| Discount factors for 5%:                  | 1.000          | 0.952         | 0.907         | 0.864         | 0.823         | 0.784         |                |
| Present value of annual cost:             | \$ (20,000.00) | \$ (4,761.90) | \$ (4,535.15) | \$ (4,319.19) | \$ (4,113.51) | \$ (3,917.63) |                |
| Total present value of life time cost:    |                |               |               |               |               |               | \$ (41,647.38) |
| Benefit derived from operational cost:    | \$ 20,000.000  | \$ 40,000.00  | \$ 40,000.00  | \$ 37,000.00  | \$ 36,000.00  | \$ 34,000.00  |                |
| Discount factors for 5%:                  | 1.00           | 0.95          | 0.91          | 0.86          | 0.82          | 0.78          |                |
| Present value of annual benefits:         | \$ 20,000.00   | \$ 38,095.24  | \$ 36,281.18  | \$ 31,961.99  | \$ 29,617.29  | \$ 26,639.89  |                |
| Total present value of lifetime benefits: |                |               |               |               |               |               | \$ 182,595.59  |

**Net present value of this alternative:** **\$ 140,948.20**

## Net Present value Analysis for Candidate 2

| Cash flow description                     | Year 0         | Year 1        | Year 2        | Year 3        | Year 4        | Year 5        | Total          |
|-------------------------------------------|----------------|---------------|---------------|---------------|---------------|---------------|----------------|
| Development cost:                         | \$ (35,000.00) |               |               |               |               |               |                |
| Operation & Maintenance cost:             |                | \$ (5,500.00) | \$ (5,500.00) | \$ (5,500.00) | \$ (5,500.00) | \$ (5,500.00) |                |
| Discount factors for 5%:                  | 1.000          | 0.952         | 0.907         | 0.864         | 0.823         | 0.784         |                |
| Present value of annual cost:             | \$ (35,000.00) | \$ (5,238.10) | \$ (4,988.66) | \$ (4,751.11) | \$ (4,524.86) | \$ (4,309.39) |                |
| Total present value of life time cost:    |                |               |               |               |               |               | \$ (58,812.12) |
| Benefit derived from operational cost:    | \$ -           | \$ 40,500.00  | \$ 40,500.00  | \$ 37,450.00  | \$ 36,425.00  | \$ 34,375.00  |                |
| Discount factors for 5%:                  | \$ 1.00        | \$ 0.95       | \$ 0.91       | \$ 0.86       | \$ 0.82       | \$ 0.78       |                |
| Present value of annual benefits:         |                | \$ 38,571.43  | \$ 36,734.69  | \$ 32,350.72  | \$ 29,966.94  | \$ 26,933.71  |                |
| Total present value of lifetime benefits: |                |               |               |               |               |               | \$ 164,557.49  |
| Net present value of this alternative:    |                |               |               |               |               |               | \$ 105,745.37  |

## **Systems Proposal**

### **I. Introduction**

This report contains the details and analysis that were involved in choosing our final course of action. Our Job was to develop a system for measuring Marine propellers electronically.

The use of electronic measurement of propellers is not new. Over 17 years ago, an Australian gentleman, was granted a patent for the electronic measurement of marine propellers. Today, the instruments that he produced are popular, but are beyond what the average marine propeller repair shop can afford. As this patent is now expired we intend to develop and manufacture such an instrument. The Australian unit sells for \$50K+ , we are aiming for \$6-7K. The instrument uses encoders to communicate its incremental travel in each axis. The software must track the changes in rotation and change in vertical position and translate them into pitch and rake values. Then the scanned values are compared to the accepted tolerances, as outlined in the ISO standard 484/2. This is done to determine whether the prop is accurate enough to fall into a specific class. Great consideration will be needed to make the software friendly to the shop floor level user.

Section I of this report contains introduction, purpose of the report, the background, and scope of this project. Section II contains how the solution was generated and feasibility analysis. Section III consists of information systems requirements. Section IV consists of the alternative solution and feasibility analysis. Section V consists of Recommendations.

### **II. Tools and techniques used**

In our search for solutions we tried to balance cost per unit with cost of development, and the opportunity cost of time taken to get the product to market. We pursued a variety of encoder packages and the options for tracking the encoder outputs. We looked at two variations in several areas. We looked at C++ verses VB. We looked at buying the latest USB encoder interface verses

working with counter chips and the parallel port. We looked at a couple of plotting and graphing COMs.

To test the feasibility of each alternative, we charted the feasibility matrix. Both candidates will fully support user-required functionality. Candidate 1 would take about 6 to 8 months to implement and the candidate 2 will take 9 to 12 months. We also charted the Net Present Value Analysis over 5 years. Net Present Value Analysis for the candidate 1 was \$140,948.20 and for the candidate 2 was \$105,745.37.

### **III. Information systems requirements**

The system requires Windows 98 or Windows 2000. We have chosen MS Access for our data management because it is known to work well with Excel and other MS products. This will allow the end user to manipulate the data that he/she wants.

### **IV. Alternative solutions and feasibility analysis**

Even though candidate 2 takes longer than candidate 1, we decided to go with candidate 2. The candidate 2 costs less per unit, and its language choice is MS Visual C++, which is more advanced than VB.

### **V. Recommendations**

It is strongly recommend that the choice will be candidate 2. Candidate 2 fully supports the requirements that are needed. It uses Visual C++, so it would be more attractive to other programmers in the future should it needed. It will use the latest USB encoder interface. Even though it will take a little longer, the payback would be great once it hits the market.

| Characteristics                                                                                                                                                                                                                                                                        | Candidate 1                                                                                         | Candidate 2                                                                                                          | Candidate 3 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Portion of System Computerized</b><br><br>Brief description of that portion of the system that would be computerized in this candidate.                                                                                                                                             | All                                                                                                 | All                                                                                                                  |             |
| <b>Benefits</b><br><br>Brief description of the business benefits that would be realized for this candidate.                                                                                                                                                                           | This solution reduces development time and development expense. Time to market is critical.         | This solution would produce a product with a lower piece price. However it would be much more expensive to implement |             |
| <b>Servers and Workstations</b><br><br>A description of the servers and workstations needed to support this candidate.                                                                                                                                                                 | MS Windows 98 or 2000                                                                               | MS Windows 98 or 2000                                                                                                |             |
| <b>Software Tools Needed</b><br><br>Software tools needed to design and build the candidate (e. g., database management system, emulators, operating systems, languages, etc.). Not generally applicable if applications software packages are to be purchased.                        | MS Visual Basic 6.0<br>Proessential for graphics<br>USB Encoder Interface<br>MS Access for Database | MS Visual C++ 6.0<br>Igrafx for graphics<br>MS Access for database                                                   |             |
| <b>Application Software</b><br><br>A description of the software to be purchased, built, accessed, or some combination of these techniques.                                                                                                                                            | Custom Solution                                                                                     | Custom solution                                                                                                      |             |
| <b>Method of Data Processing</b><br><br>Generally some combination of: on-line, batch, deferred batch, remote batch, and real-time.                                                                                                                                                    | Real-time                                                                                           | Real-time                                                                                                            |             |
| <b>Output Devices and Implications</b><br><br>A description of output devices that would be used, special output requirements, (e.g. network, preprinted forms, etc.), and output considerations (e.g., timing constraints).                                                           | Monitors and Printers                                                                               | Monitors and Printers                                                                                                |             |
| <b>Input Devices and Implications</b><br><br>A description of Input methods to be used, input devices (e.g., keyboard, mouse, etc.), special input requirements, (e.g. new or revised forms from which data would be input), and input considerations (e.g., timing of actual inputs). | Keyboard, Mouse, Encoders, and Switches on the Pitch gage                                           | Keyboard, Mouse, Encoders and Switches on the Pitch gage                                                             |             |
| <b>Storage Devices and Implications</b><br><br>Brief description of what data would be stored, what data would be accessed from existing stores, what storage media would be used, how much storage capacity would be needed, and how data would be organized.                         | Hard drive and discs                                                                                | Hard drive and discs                                                                                                 |             |

| Feasibility Criteria                                                                                                                                                                                                                                                                                                                       | Wt.  | Candidate 1                                                                                                                                                                                                                                                                                                                                                                                 | Candidate 2                                                                                                                                                                                                                                                                                                 | Candidate 3 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Operational Feasibility</b><br><br><b>Functionality.</b> A description of to what degree the candidate would benefit the organization and how well the system would work.<br><br><b>Political.</b> A description of how well received this solution would be from both user management, user, and organization perspective.             | 30%  | Fully supports user required functionality.<br><br><br>Score: 100                                                                                                                                                                                                                                                                                                                           | Fully supports user required functionality.<br><br><br>Score: 100                                                                                                                                                                                                                                           | Score:      |
| <b>Technical Feasibility</b><br><br><b>Technology.</b> An assessment of the maturity, availability (or ability to acquire), and desirability of the computer technology needed to support this candidate.<br><br><b>Expertise.</b> An assessment to the technical expertise needed to develop, operate, and maintain the candidate system. | 30%  | Current system in measuring Marine Propellers is by hand. This system will replace the current hand system with computerized system. However, there will one extra step because this system does not have encoder already built in.<br><br>It will take less time to implement using Visual Basic 6.0 because it is simpler language, but it is not as popular as C++.<br><br><br>Score: 85 | This system will also replace the current hand measuring system with computerized system. The transition will be very easy.<br><br>It will take longer to implement with MS visual C++ because it is harder and more complex language. However, it is more popular language than MSVB.<br><br><br>Score: 95 | Score:      |
| <b>Economic Feasibility</b><br><br><b>Cost to develop:</b><br><br><b>Payback period (discounted):</b><br><br><b>Net present value:</b><br><br><b>Detailed calculations:</b>                                                                                                                                                                | 30%  | Approximately \$20,000.00<br><br>Approximately 6 months<br><br>Approximately \$140948.00<br><br>Score: 95                                                                                                                                                                                                                                                                                   | Approximately \$35,000.00<br><br>Approximately 10.9 months<br><br>Approximately \$ 105745.00<br><br>Score: 90                                                                                                                                                                                               | Score:      |
| <b>Schedule Feasibility</b><br><br>An assessment of how long the solution will take to design and implement.                                                                                                                                                                                                                               | 10%  | 6 months<br><br>Score: 95                                                                                                                                                                                                                                                                                                                                                                   | 9 to 12 months<br><br>Score: 90                                                                                                                                                                                                                                                                             | Score:      |
| <b>Ranking:</b>                                                                                                                                                                                                                                                                                                                            | 100% | 93.5                                                                                                                                                                                                                                                                                                                                                                                        | 94.5                                                                                                                                                                                                                                                                                                        |             |

Customer Name: John Doe  
 Vessel Name: Fish Killer  
 Number of Blades: 3

Marked Diameter: 28  
 Marked Pitch: 32  
 Rotation: R

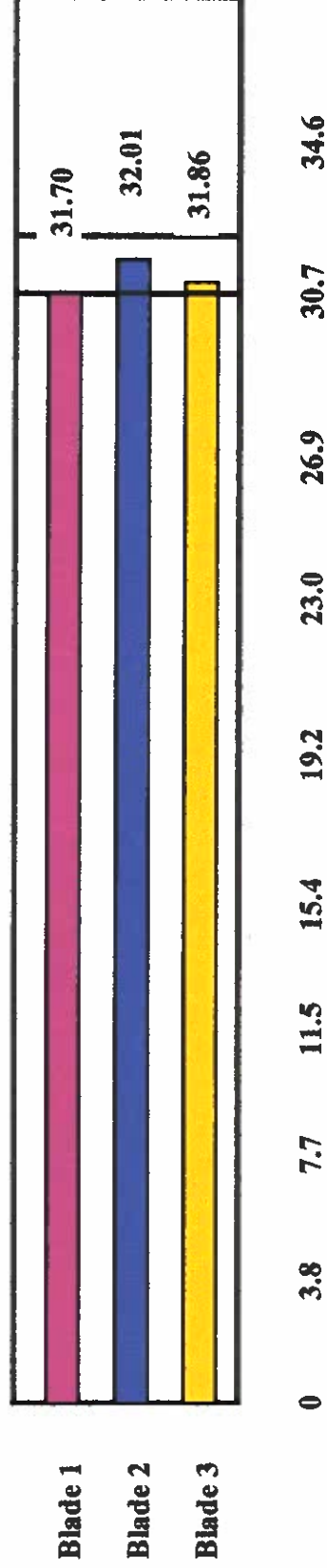
Measured Diameter: 28.00  
 Measured Pitch: 31.856  
 Class: I

Radii Averages:

| r/R  | Bld 1 | Bld 2 | Bld 3 |
|------|-------|-------|-------|
| 50 % | 31.56 | 31.84 | 31.57 |
| 70 % | 31.73 | 32.02 | 31.84 |
| 90 % | 31.80 | 32.17 | 32.17 |

Blade Averages:

| Avg. | 31.697 | 32.010 | 31.860 |
|------|--------|--------|--------|
|------|--------|--------|--------|





# EXAMPLE OUTPUT

Customer Name: John Doe  
Vessel Name: Fish Killer  
Number of Blades: 3

Marked Diameter: 28  
Marked Pitch: 32  
Rotation: R

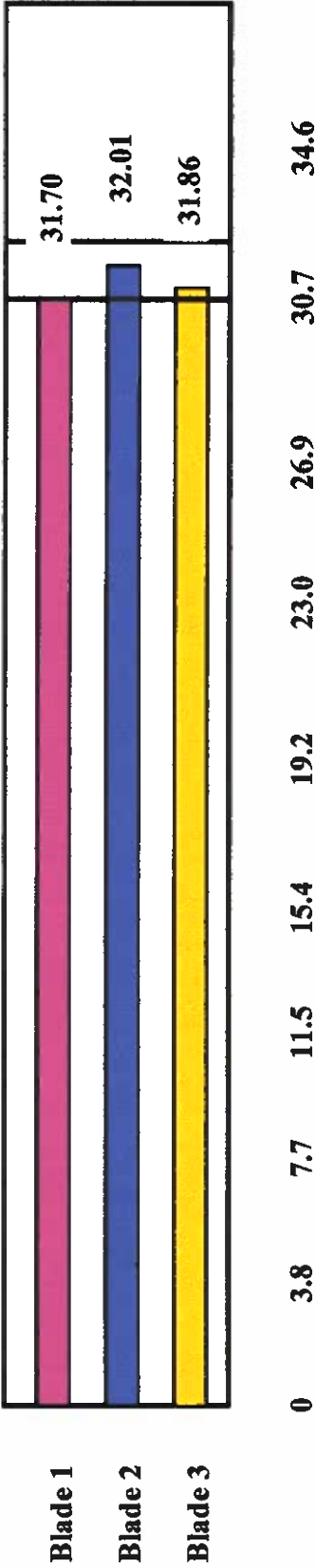
Measured Diameter: 28.00  
Measured Pitch: 31.856  
Class: I

Radii Averages:

| r/R  | Bld 1 | Bld 2 | Bld 3 |
|------|-------|-------|-------|
| 50 % | 31.56 | 31.84 | 31.57 |
| 70 % | 31.73 | 32.02 | 31.84 |
| 90 % | 31.80 | 32.17 | 32.17 |

Blade Averages:

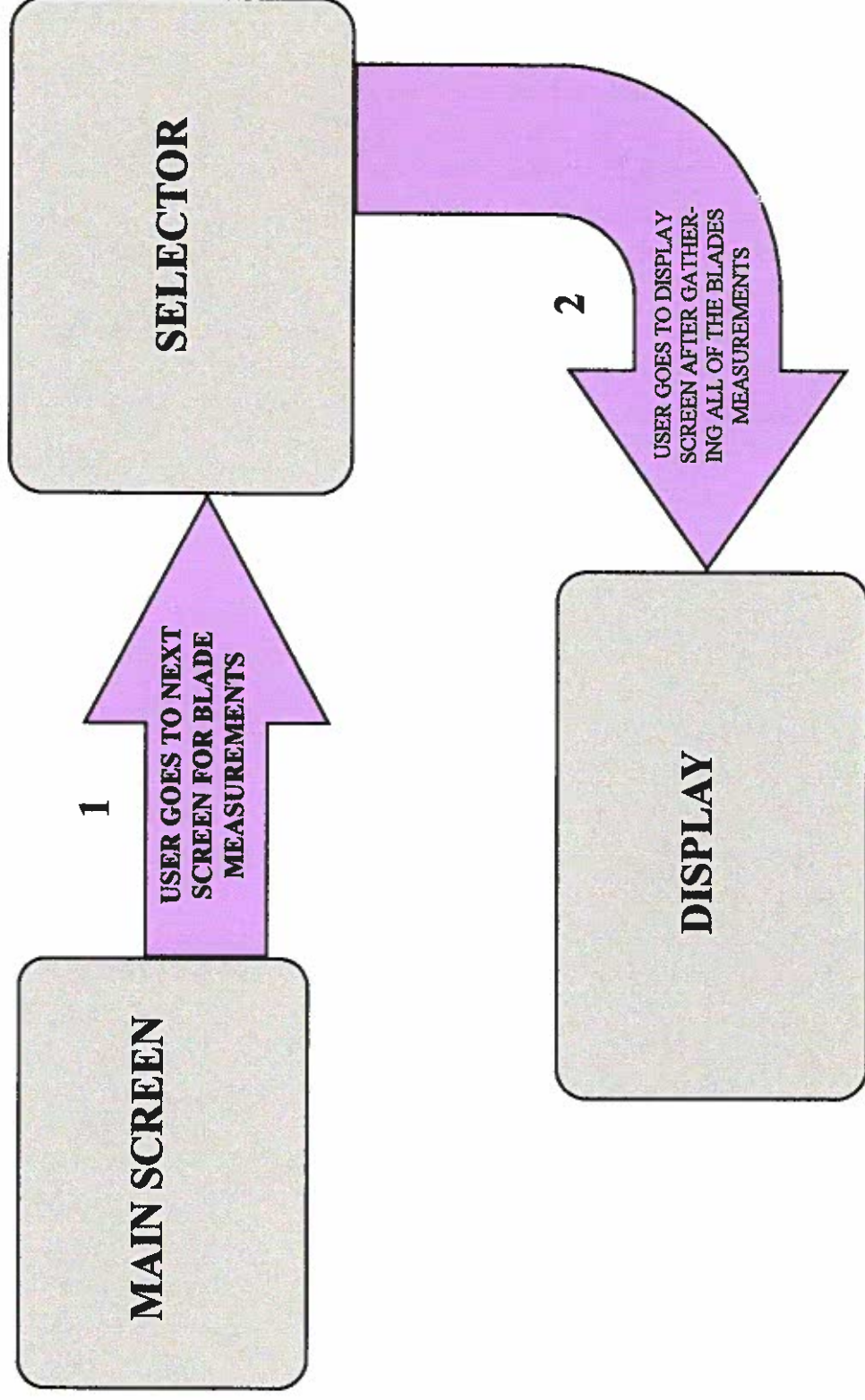
|      |        |        |        |
|------|--------|--------|--------|
| Avg. | 31.697 | 32.010 | 31.860 |
|------|--------|--------|--------|



## **COMPLETE USER DIALOG**

- 1. Zero out all the axis.**
- 2. Start measuring one blade.**
- 3. Go to next blade after all the measurements are done on first blade.**
- 4. Repeat the measurement process until all the blades are measured.**
- 5. Plot.**

# STATE TRANSITION DIAGRAM

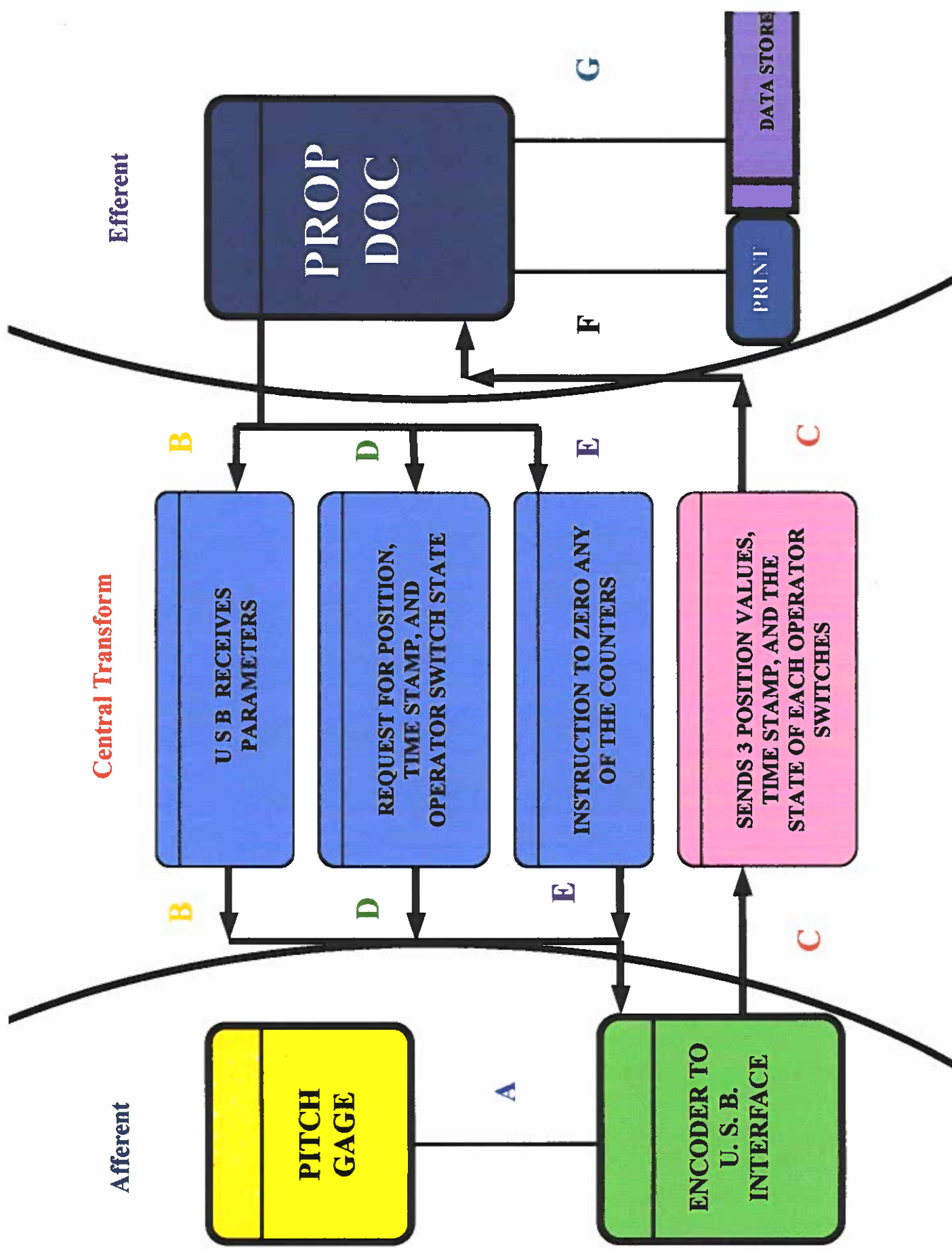


# **MILESTONE**

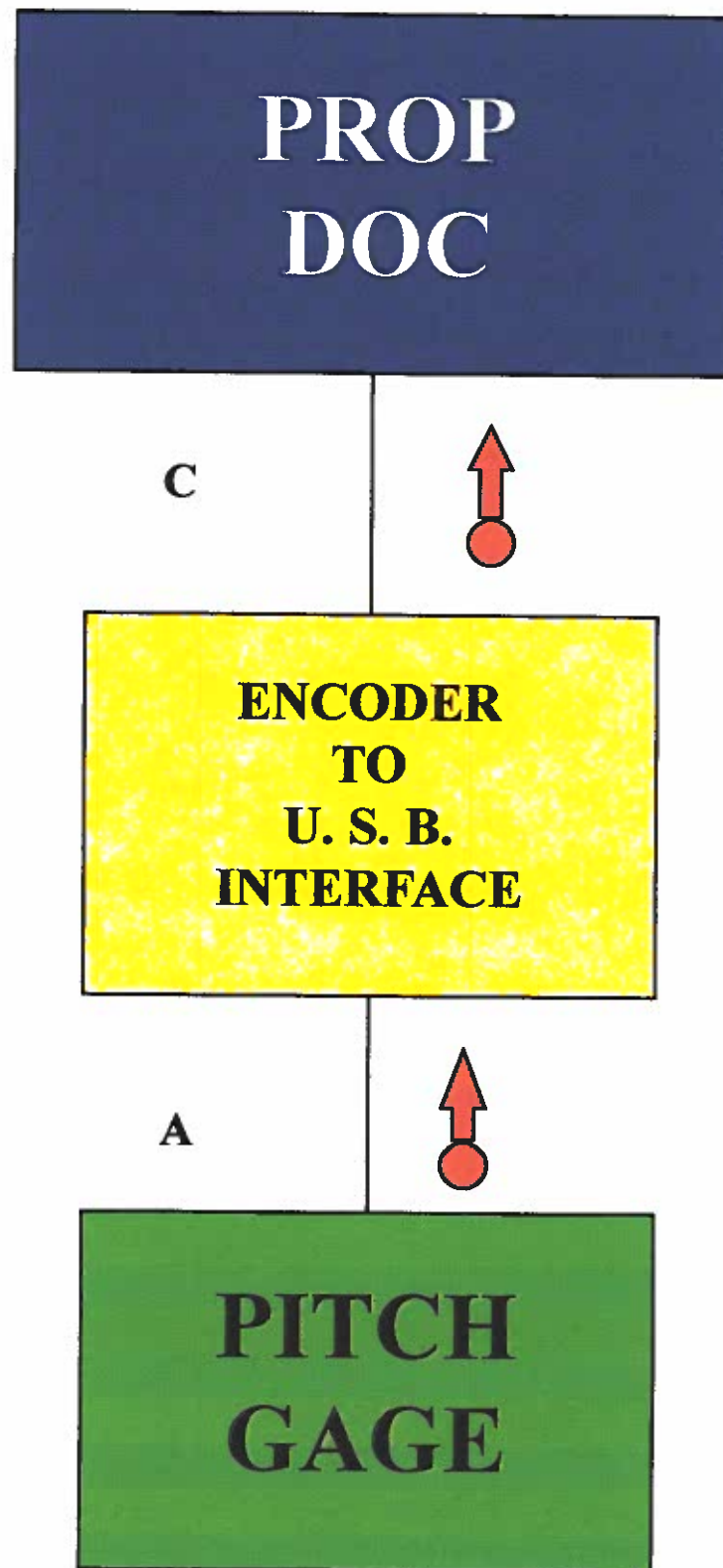
**11**

**BY  
Y. M. INC.**

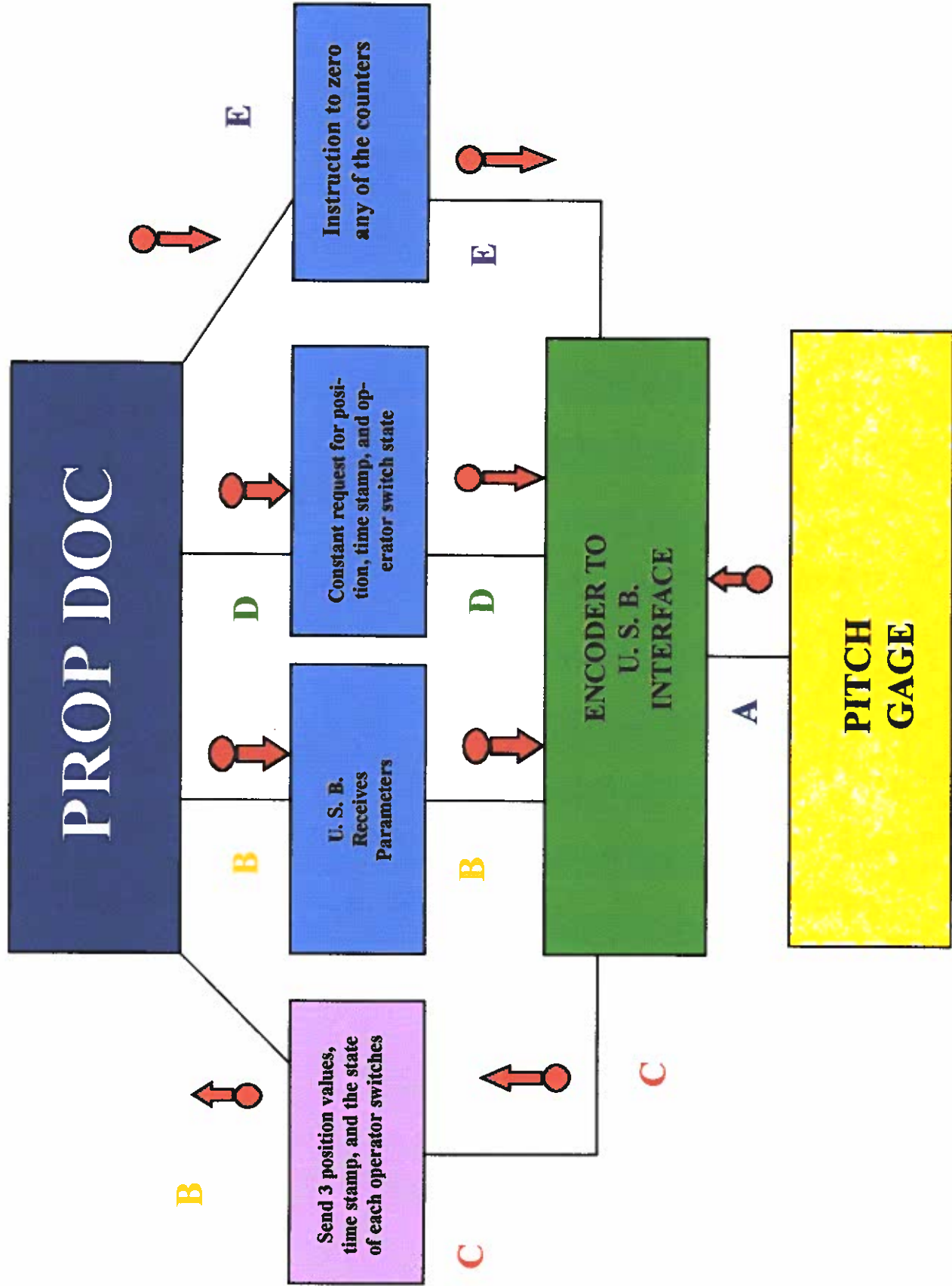
# PARTITIONED DFD



# AFFERENT STRUCTURE CHART

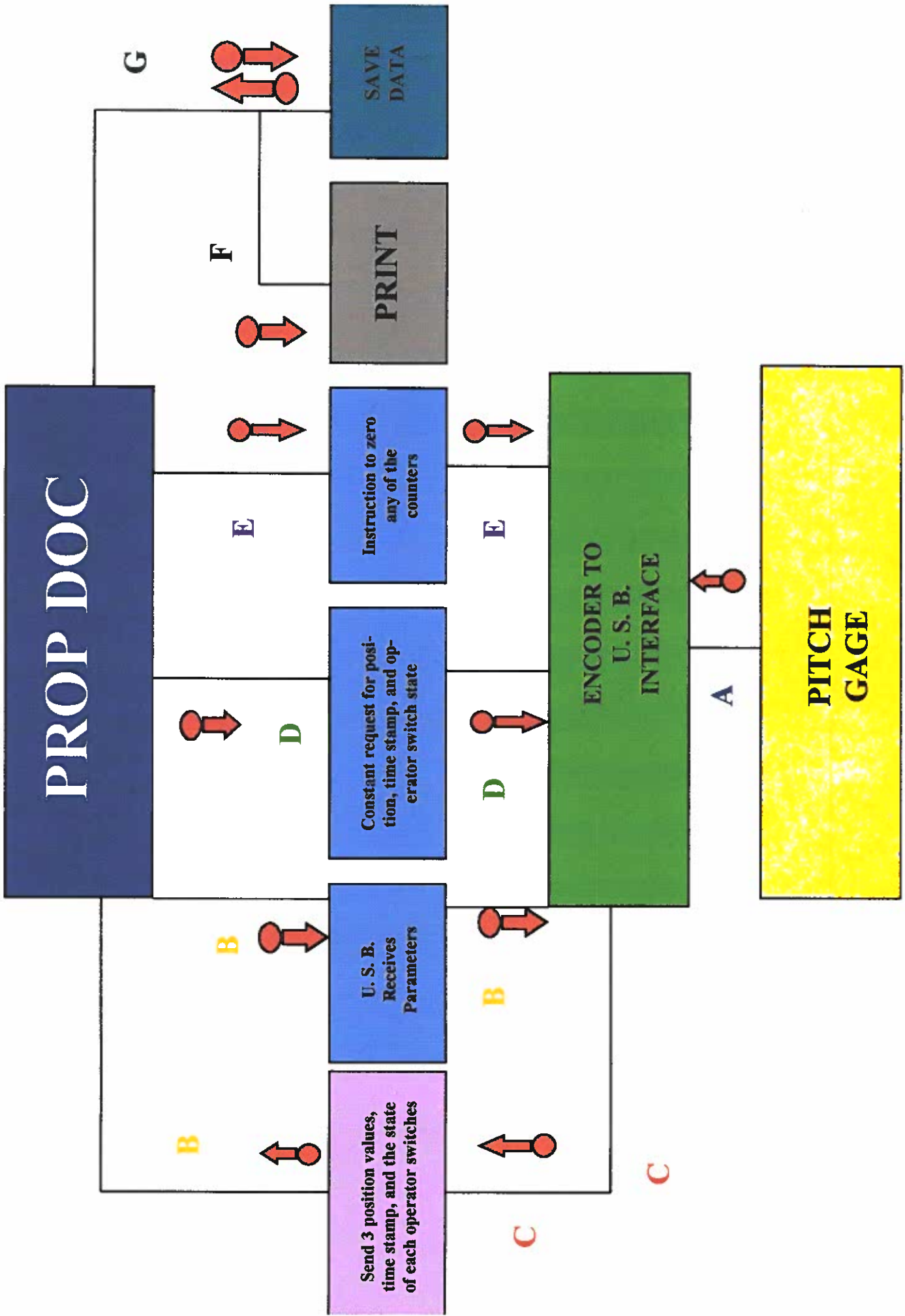


# CENTRAL TRANSFORM STRUCTURE CHART





# EFFERENT STRUCTURE CHART





# **MILESTONE**

**12**

**BY  
Y. M. INC.**

**Y. M. INC.**

**1234 Yons Street.**

**N. CHARLESTON, SC 29418**

**843-767-8643 FAX 843-767-9181**

**GENERAL SYSTEM REPORT**

**BY**

**Y. M. INC.**

**FOR**

**PESCO, INC.**

**11-27-00**

Mr. Michael Meyer  
3333 North American Street  
North Charleston, SC 29418

December 4, 2000

Y. M. Inc.  
1234 Yons Road  
Charleston, Sc 29418

Dear Mr. Meyer,

Y. M. Inc is very pleased to inform you and your associates that we have completed the installation of a custom designed computer based system, Prop DOC. Y. M. Inc. has designed the system with Pesco's specific needs in mind. The Prop DOC, through the hard work of our associates, has been implemented to be bug free and very user friendly.

The Prop DOC has the capability to automate the marine propeller measuring process that was implemented manually in the past. The Prop DOC will help reduce excessive steps, efforts, and time. Through proper maintenance and future appropriate upgrades, your new system will provide Pesco, Inc. with many years of dependable service.

All objectives have been achieved within the specified schedule and within the budget constraints set forth by our contract. We are confident that you will be satisfied with the product and our service. We stand behind all of our products and services. Please review our attached report addressing the project in detail. If you have any questions, please do not hesitate to call me personally, or any of our associates, here at Y. M. Inc.

We at Y. M. Inc. have enjoyed working with you and your staff. We are also looking forward to a continued business relationship in the future.

Sincerely,

*Yon Meyer*

Yon Meyer  
President, Y. M. Inc.

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## Introduction

The purpose of this report is to convey the results of the design and implementation of the marine propeller measuring system to Pesco, Inc. Both management and end users will find this report beneficial to help understand the related issues involved in the design and implementation.

The usefulness of our service became apparent after meeting with the president of Pesco, Inc. On September thirteenth of this year, we asked to quote the installation of a new system for Pesco, Inc. We were afforded the opportunity to commence with the installation at that time.

On September eighteenth, we submitted a time frame for the installation of a system of our design. The estimated time span of our system planning to implementation was laid out for you.

The manual marine propeller measuring system was studied. Our staff, with cooperation from you and your staff, became very familiar with your procedures. We studied closely the need for automated system. We submitted on September twenty-fourth and twenty-eighth, to Pesco, Inc. our diagram showing how the new processes will interact with each other. Included were context diagram, event decomposition diagram, system diagram, and physical data flow diagram. Upon arrival of these diagrams, we submitted, on October fifth, location connectivity diagram showing how the different location of Pesco, Inc. would integrate by the system.

On October twelfth we moved on to detailing the individual processes and the events that triggered each process. Included was data structure of inputs and outputs.

Your system can now be used for the measuring marine propellers and generate reports automatically. Pesco, Inc. will realize an increased accuracy and less time consuming.

The above synopsis gives a quick look at the implementation process from the start of this project to present day. Also, you will find several steps outlining other steps of our process. These include the **standard and specifications** describing the hardware and software used in the system. A **system overview**, providing diagrams and flow charts showing the different processes and their relationship to each other. **The File and Data specifications**, details the structure for database. **Program specifications** helping the users understand the system and how it works. A detail of the **implementation schedule used to implement the entire project**.

## **Standards and Specifications**

**The new system will require Windows 95 or higher operating system. Pesco, Inc. has Windows 98 operating systems so buying a new computer was unnecessary. The computer that Pesco, Inc. currently has, ten gigabytes of hard drive, and sixteen megabyte of RAM. This is plenty more than what is required. The new system only required one gigabyte of hard drive and sixteen megabytes of RAM.**

**However, Pesco, Inc. will need to invest in an online service in order to upgrade via e-mail. The president of Pesco, Inc. is fully aware of this matter.**

**The database functions, will be handled by the software produced by Y. M. Inc.**

## **System Overview**

We are providing four different diagrams: one context diagram, one event decomposition diagram, one physical data flow diagram, and one partitioned DFD.

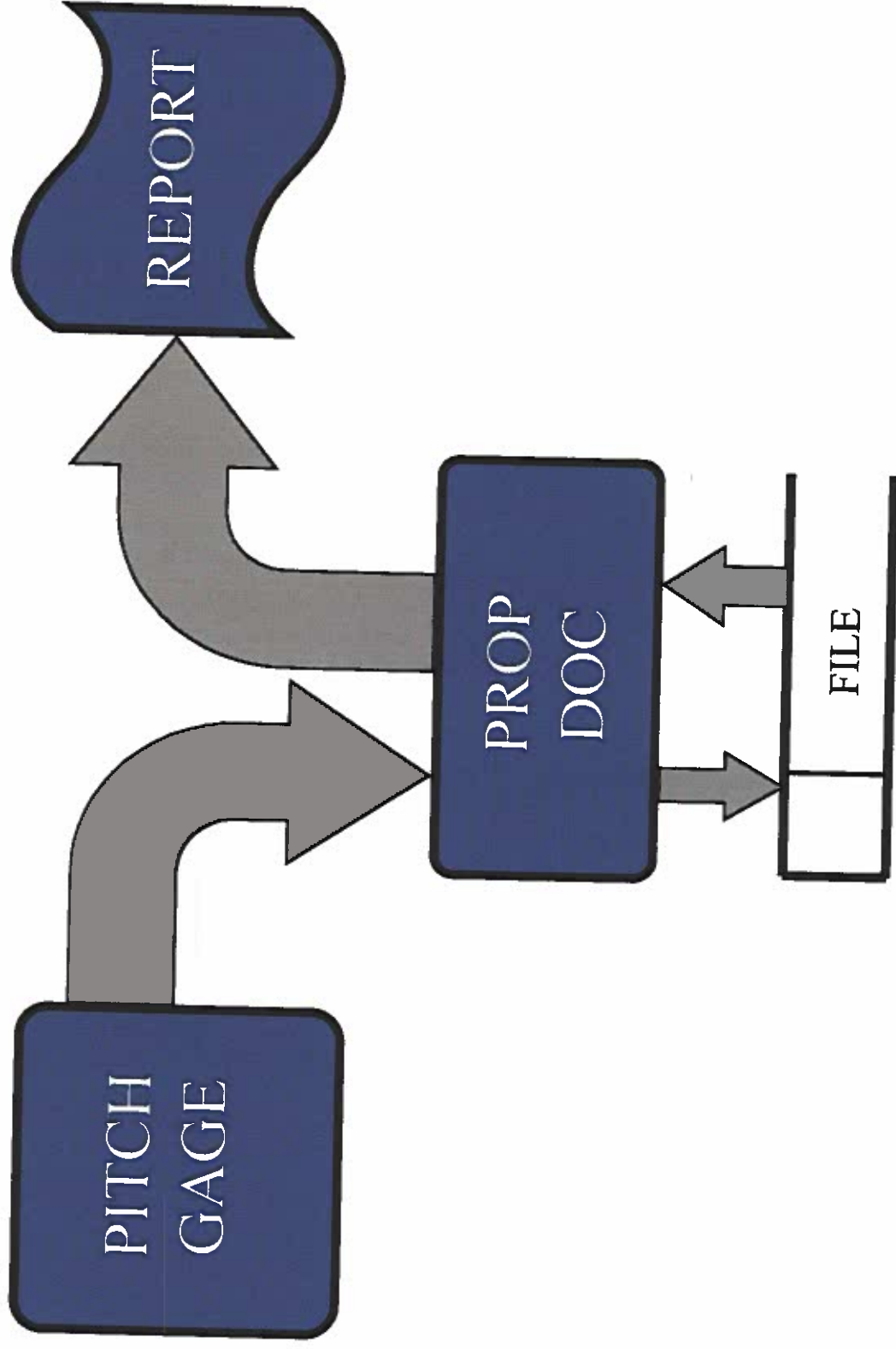
Unlike the many other system, this system is heavy on processes instead of data exchange. The context diagram consists of only external agent, and data storage.

The event decomposition diagram contains processes only. It is a hierarchy chart of the processes in the information service system. Each processes is broken down into smaller processes, until the processes could not be broken down anymore.

The physical data flow diagram shows how the data flows between processes, external agent and the data stores. The arrows represent the direction of flows .

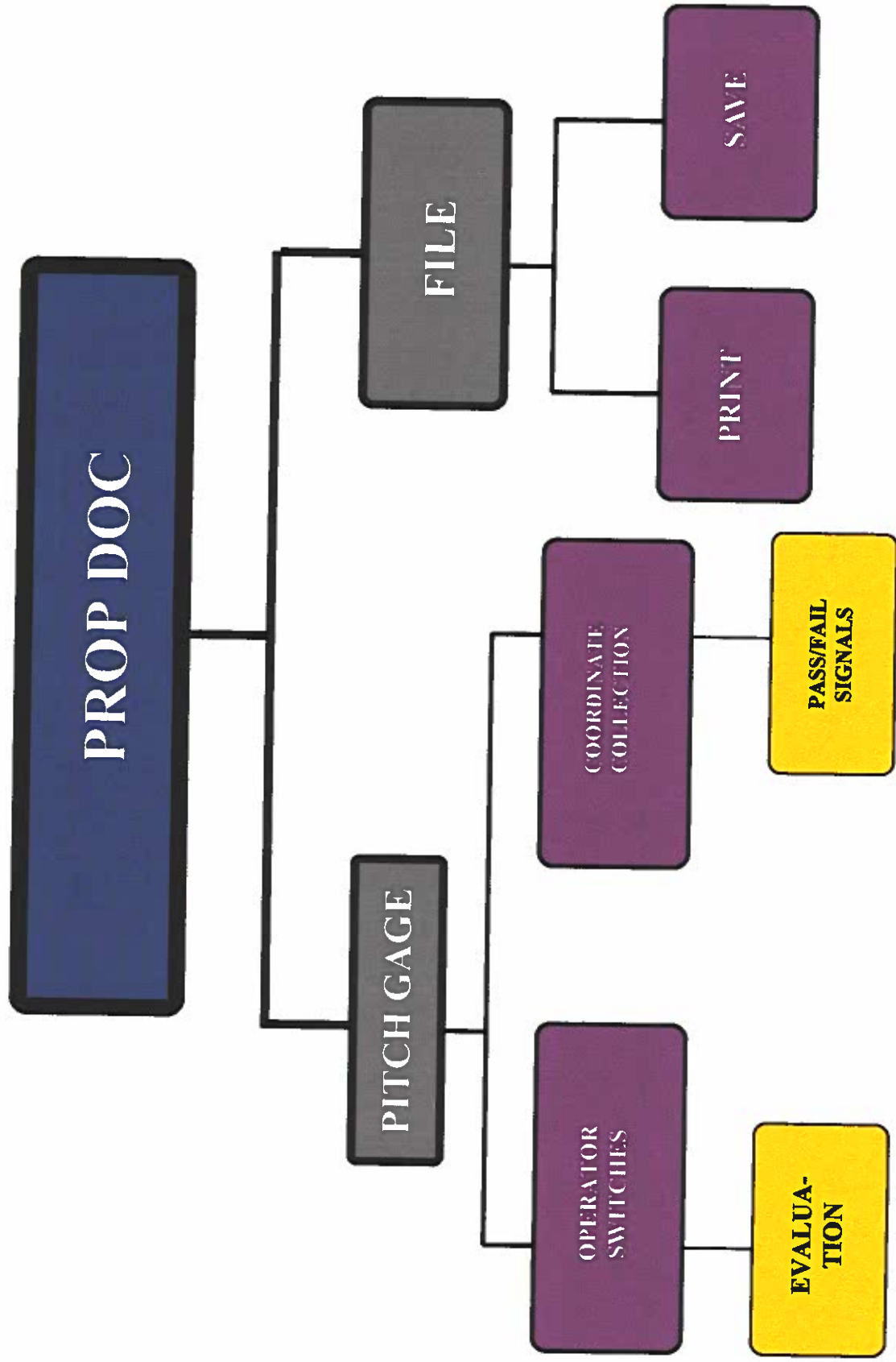


# CONTEXT DIAGRAM

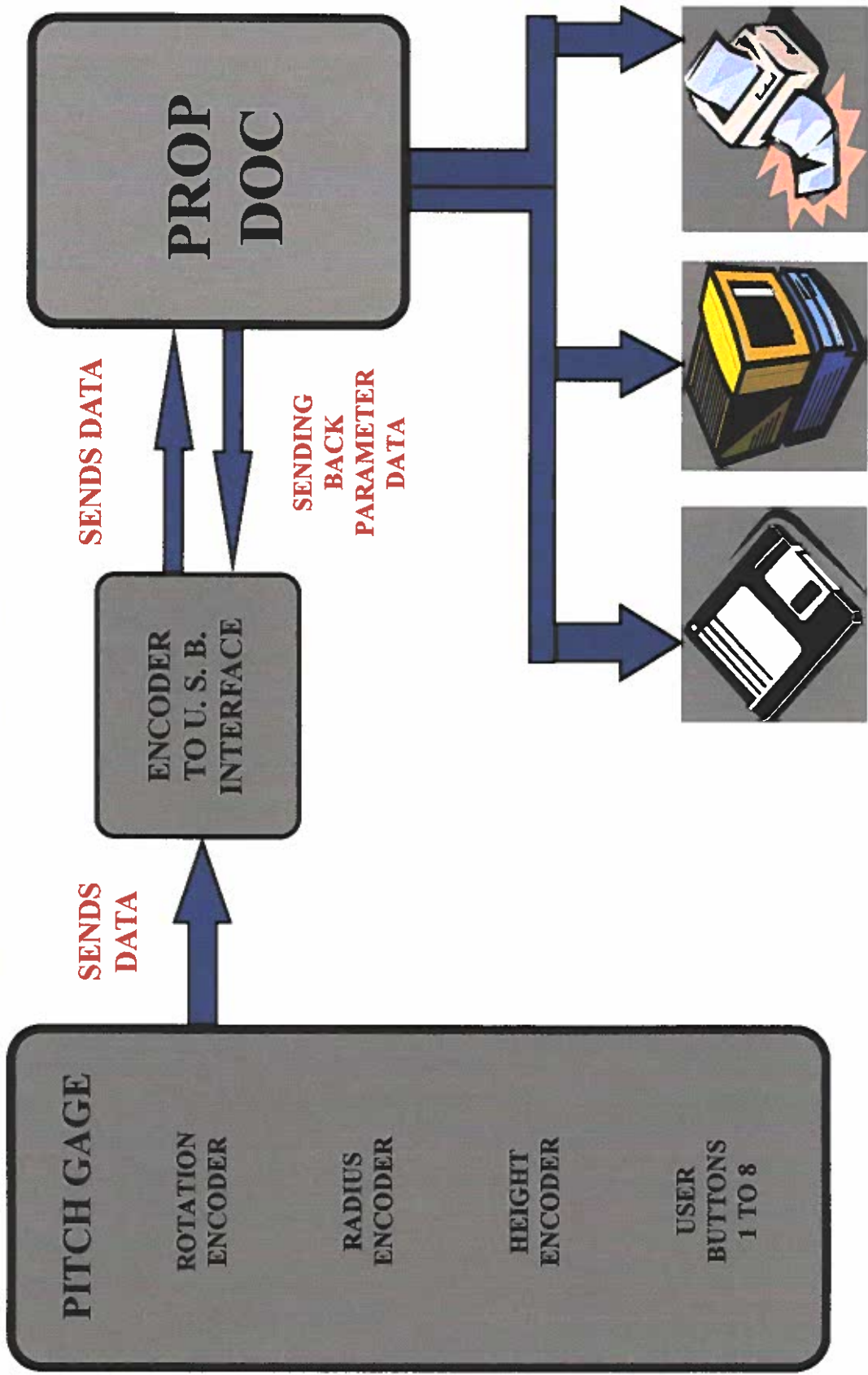




# EVENT DECOMPOSITION DIAGRAM



# PHYSICAL DATA FLOW DIAGRAM



## **File Specifications**

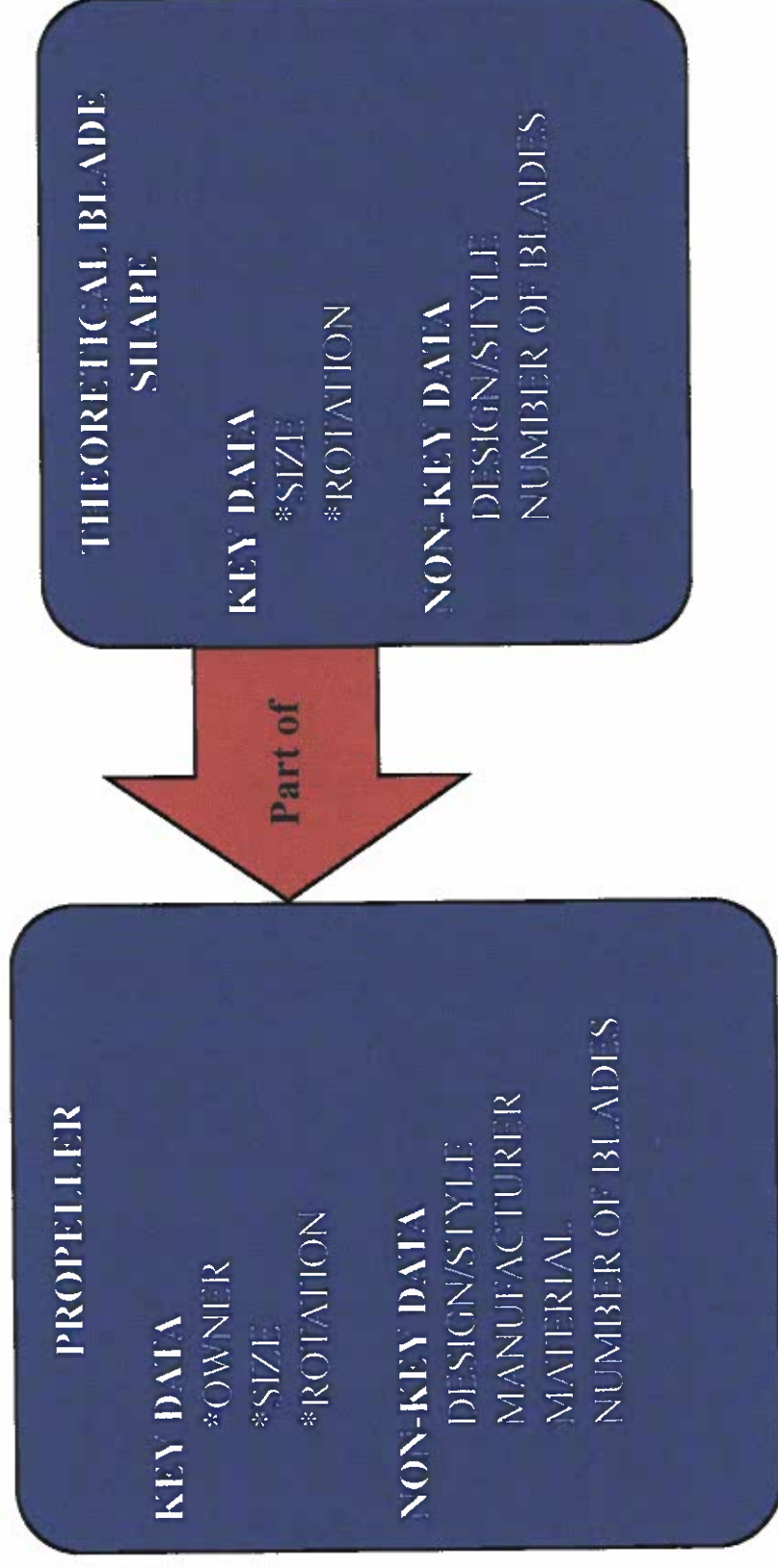
This section details the files used within the new system.

All the propeller files, must consist of owner's name, design/style, propeller size, manufacturer, Rotation, and the materials that are required by the system.

Theoretical file blade shape file must consist of size, rotation, design/style, and number of blades.

# Logical Data Model In 3rd Normal Form

\* PRIMARY KEY



## **Program Specifications**

This section consists of technical documentation required for Pesco, Inc. to develop the new system. It contains:

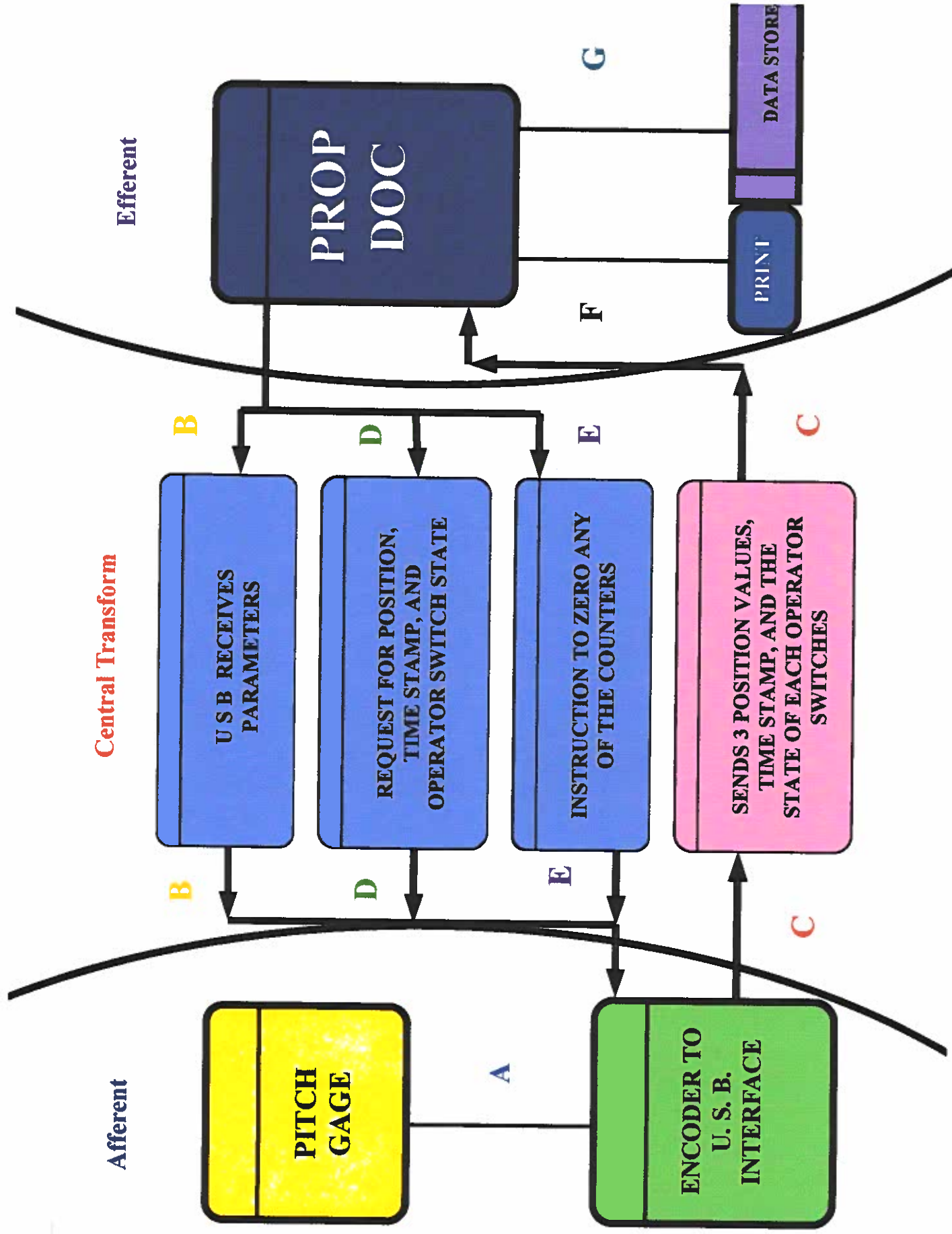
Partitioned DFD

Terminal screen display layout charts

Input/output project repositories

Decision Tables.

# PARTITIONED DFD

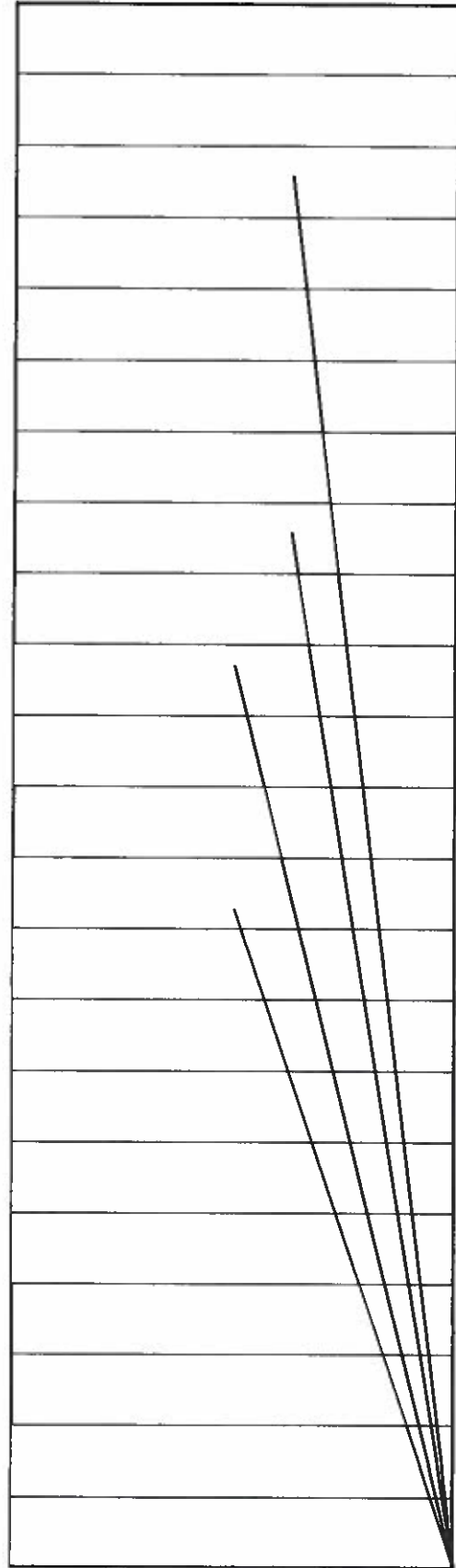




PROCESSING



STOP



# Prop D O C

Enter the following items for each propellers. Use tab or arrow keys to move from item to item.

|                              |                          |
|------------------------------|--------------------------|
| Date: 11-10-00               | File name: ScanData1.txt |
| Customer Name: John Doe      | Vessel Name: Fish Killer |
| Manufacturer: Michigan Wheel | Material: Nibral         |
| Design/Style: Dyna Jet       | Bore: 2.0                |
| Class: I                     | DAR: .55                 |
| Cupping: 2 --.033            | Rotation: R              |
| Marked Diameter: 28          | Measured Diameter: 28.00 |
| Marked Pitch: 32             | Measured Pitch: 31.856   |

Push enter when you are finished.



# Prop D O C

Enter the following items for each propellers. Use tab or arrow keys to move from item to item.

|                  |                    |
|------------------|--------------------|
| Date:            | File name:         |
| Customer Name:   | Vessel Name:       |
| Manufacturer:    | Material:          |
| Design/Style:    | Bore:              |
| Class:           | DAR:               |
| Cupping:         | Rotation:          |
| Marked Diameter: | Measured Diameter: |
| Marked Pitch:    | Measured Pitch:    |

Push enter when you are finished.

Customer Name: John Doe  
Vessel Name: Fish Killer  
Number of Blades: 3

Marked Diameter: 28  
Marked Pitch: 32  
Rotation: R

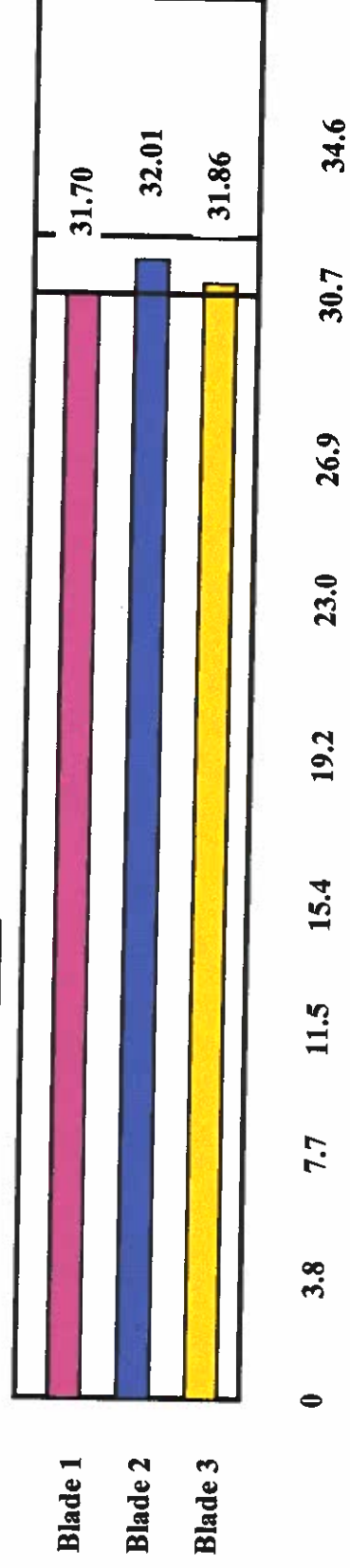
Measured Diameter: 28.00  
Measured Pitch: 31.856  
Class: I

Radii Averages:

| r/R  | Bld 1 | Bld 2 | Bld 3 |
|------|-------|-------|-------|
| 50 % | 31.56 | 31.84 | 31.57 |
| 70 % | 31.73 | 32.02 | 31.84 |
| 90 % | 31.80 | 32.17 | 32.17 |

Blade Averages:

| Avg. | 31.697 | 32.010 | 31.860 |
|------|--------|--------|--------|
|------|--------|--------|--------|



EXAMPLE OUTPUT

Customer Name: John Doe  
Vessel Name: Fish Killer  
Number of Blades: 3

Marked Diameter: 28  
Marked Pitch: 32  
Rotation: R

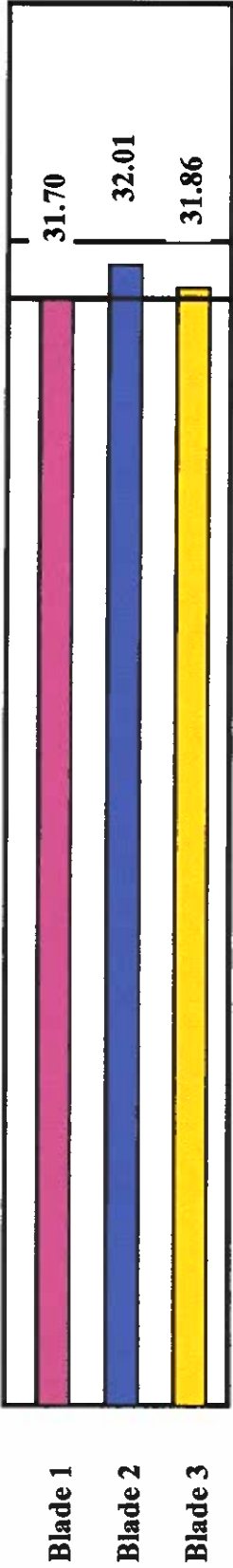
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Blade Averages:

| Avg. | 31.697 | 32.010 | 31.860 |
|------|--------|--------|--------|
|------|--------|--------|--------|



# DECISION TABLE 1

| Class S Acceptance Policy                  |   | Rules |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |
|--------------------------------------------|---|-------|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|
|                                            |   | 1     | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| Local Pitch Deviation $\leq 1.5\%$         | Y | N     | N | N | N | N | Y | Y | Y | Y | Y  | Y  | N  | Y  | N  | N  | N  |
| Mean Pitch Cords Deviation $\leq 1.0\%$    | Y | Y     | N | N | N | N | N | N | N | Y | Y  | N  | Y  | Y  | Y  | N  | Y  |
| Mean Pitch Blade Deviation $\leq .75\%$    | Y | Y     | Y | N | N | N | Y | N | N | N | Y  | Y  | N  | N  | N  | Y  | Y  |
| Mean Pitch Propeller Deviation $\leq .5\%$ | Y | Y     | Y | Y | Y | N | Y | Y | N | Y | N  | N  | Y  | N  | N  | N  | N  |
| Accept for Class S                         | X |       |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |
| Test for Next lower Class                  |   | X     | X | X | X | X | X | X | X | X | X  | X  | X  | X  | X  | X  | X  |

# DECISION TABLE 2

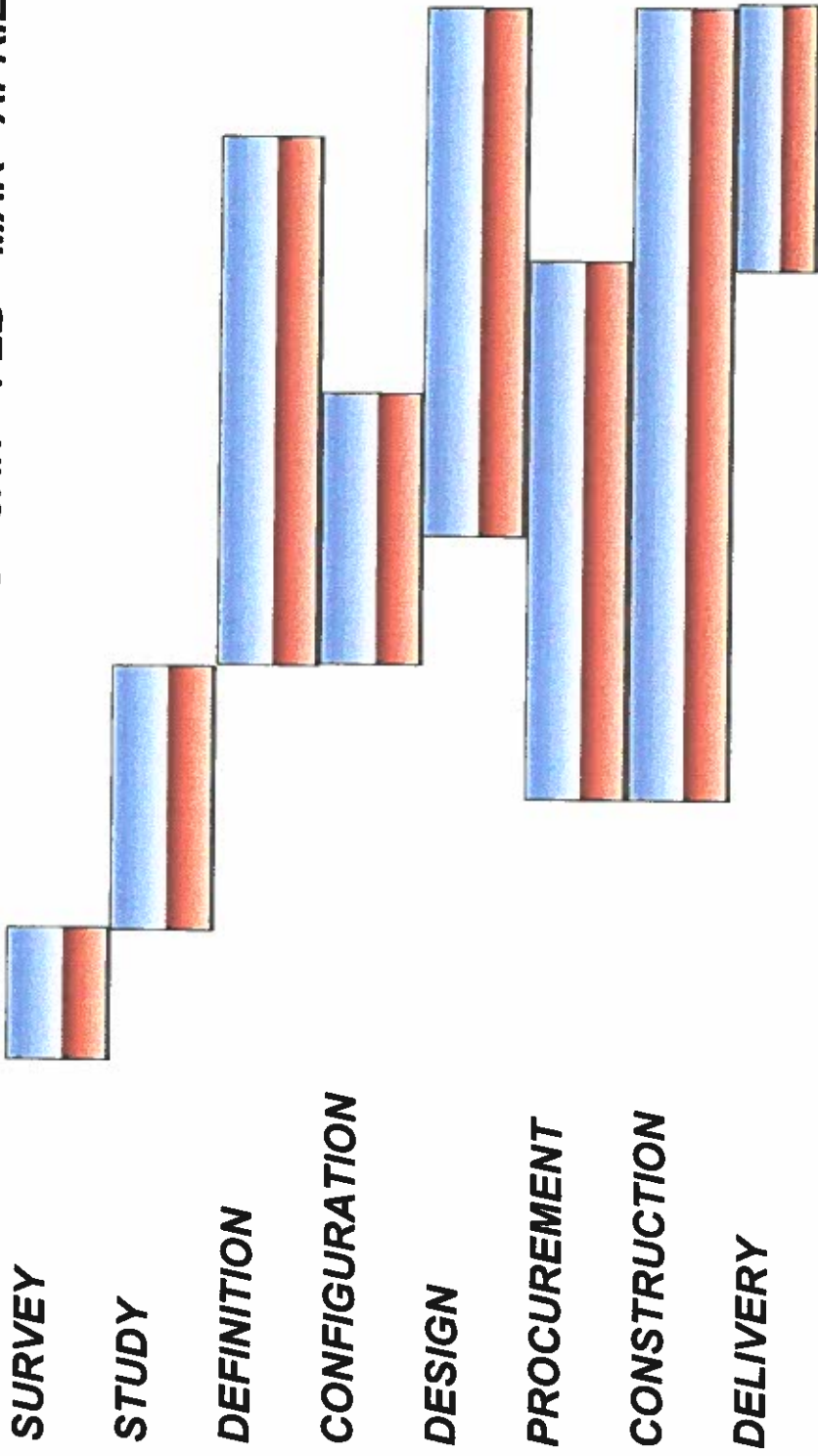
| Class 1 Acceptance Policy                   |  | Rules |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |
|---------------------------------------------|--|-------|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|
|                                             |  | 1     | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| Local Pitch Deviation $\leq 2\%$            |  | Y     | N | N | N | N | Y | Y | Y | Y | Y  | Y  | N  | Y  | N  | N  | N  |
| Mean Pitch Cords Deviation $\leq 1.5\%$     |  | Y     | Y | N | N | N | N | N | N | Y | Y  | N  | Y  | Y  | Y  | N  | Y  |
| Mean Pitch Blade Deviation $\leq 1.0\%$     |  | Y     | Y | Y | N | N | Y | N | N | Y | Y  | Y  | N  | N  | N  | Y  | Y  |
| Mean Pitch Propeller Deviation $\leq .75\%$ |  | Y     | Y | Y | Y | N | Y | Y | N | Y | N  | N  | Y  | N  | N  | N  | N  |
| Accept for Class 1                          |  | X     |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |
| Test for Next lower Class                   |  |       | X | X | X | X | X | X | X | X | X  | X  | X  | X  | X  | X  | X  |

## **Implementation Schedule**

This section contains the time frame anticipated for each phase in development of this system. We have included Gantt chart to further clarify the time line.

# SYSTEM DEVELOPMENT LIFE CYCLE

SEP OCT NOV DEC JAN FEB MAR APRIL



## **Recommendations and Conclusions**

This report present the design of new efficient marine propeller measuring system for Pesco, Inc. This system will standardize your company business and at the same time drastically reduce time and inaccuracy.

At this time, we are submitting our final proposal to implement this new system for Pesco, Inc. We anticipate a time frame of ten weeks to install the new hardware and testing of the new software. During last two weeks of this time, we will conduct training for the users from Pesco, Inc.. We anticipate continuing this training for a week longer . Our systems support will be ongoing , and we will provide additional training as needed.

We, here at Y. M. Inc. , would like to extend our gratitude to all employees of Pesco, Inc. for their cooperation in this project. We especially extend our gratitude to the management of Pesco, Inc. for giving us this opportunity to provide this service for you. We look forward to our future business relationship with Pesco, Inc.